

with *dwell*. A Geneva wheel needs a minimum of three stops to work. The maximum number of stops is limited only by the size of the wheel.

RATCHET AND PAWL Figure 2-12b shows a ratchet and pawl mechanism. The arm pivots about the center of the toothed ratchet wheel and is moved back and forth to index the wheel. The driving pawl rotates the ratchet wheel (or ratchet) in the counter-clockwise direction and does no work on the return (clockwise) trip. The locking pawl prevents the ratchet from reversing direction while the driving pawl returns. Both pawls are usually spring-loaded against the ratchet. This mechanism is widely used in devices such as "ratchet" wrenches, winches, etc.

LINEAR GENEVA MECHANISM There is also a variation of the Geneva mechanism which has linear translational output, as shown in Figure 2-12c. This mechanism is analogous to an open Scotch yoke device with multiple yokes. It can be used as an intermittent conveyor drive with the slots arranged along the conveyor chain or belt. It also can be used with a reversing motor to get linear, reversing oscillation of a single slotte, put slider.

2.11 INVERSION

It should now be apparent that there are many possible linkages for any situation. Even with the limitations imposed in the number synthesis example (one *DOF*, eight links, up to hexagonal order), there are eight linkage combinations shown in Table 2-2 (p. 36), and these together yield 19 valid isomers as shown in Table 2-3 (p. 38). In addition, we can introduce another factor, namely mechanism inversion. An inversion is *created by grounding a different link in the kinematic chain*. Thus there are as many inversions of a given linkage as it has links.

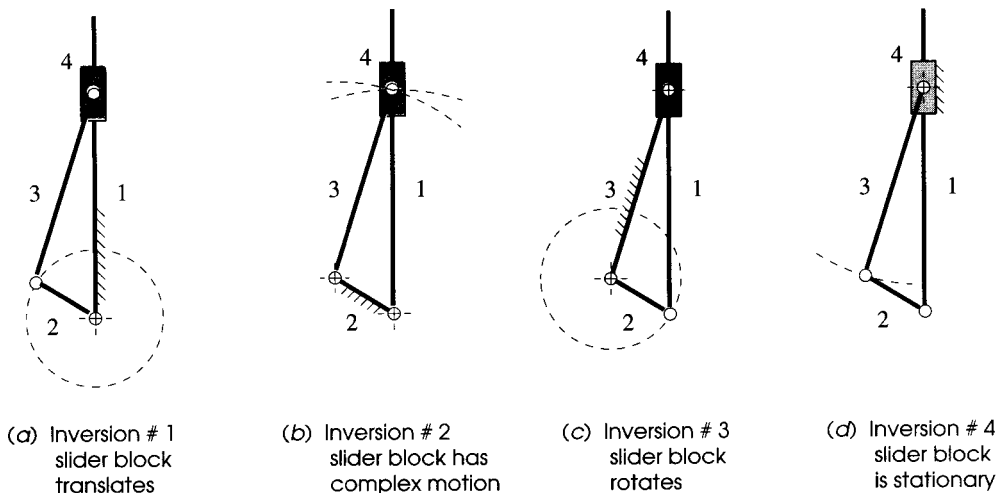


FIGURE 2-13

Four distinct inversions of the fourbar slider-crank mechanism (each black link is stationary—all red links move)

The motions resulting from each inversion can be quite different, but some inversions of a linkage may yield motions similar to other inversions of the same linkage. In these cases only some of the inversions may have distinctly different motions. We will denote the *inversions which have distinctly different motions* as distinct inversions.

Figure 2-13 (previous page) shows the four inversions of the fourbar slider-crank linkage, all of which have distinct motions. Inversion #1, with link 1 as ground and its slider block in pure translation, is the most commonly seen and is used for piston engines and piston pumps. Inversion #2 is obtained by grounding link 2 and gives the Whitworth or crank-shaper quick-return mechanism, in which the slider block has complex motion. (Quick-return mechanisms will be investigated further in the next chapter.) Inversion #3 is obtained by grounding link 3 and gives the slider block pure rotation. Inversion #4 is obtained by grounding the slider link 4 and is used in hand operated, well pump mechanisms, in which the handle is link 2 (extended) and link 1 passes down the well pipe to mount a piston on its bottom. (It is upside down in the figure.)

The Watt's sixbar chain has two distinct inversions, and the Stephenson's sixbar has three distinct inversions, as shown in Figure 2-14. The pin-jointed fourbar has four distinct inversions: the crank-rocker, double-crank, double-rocker, and triple-rocker which are shown in Figures 2-15 and 2-16.

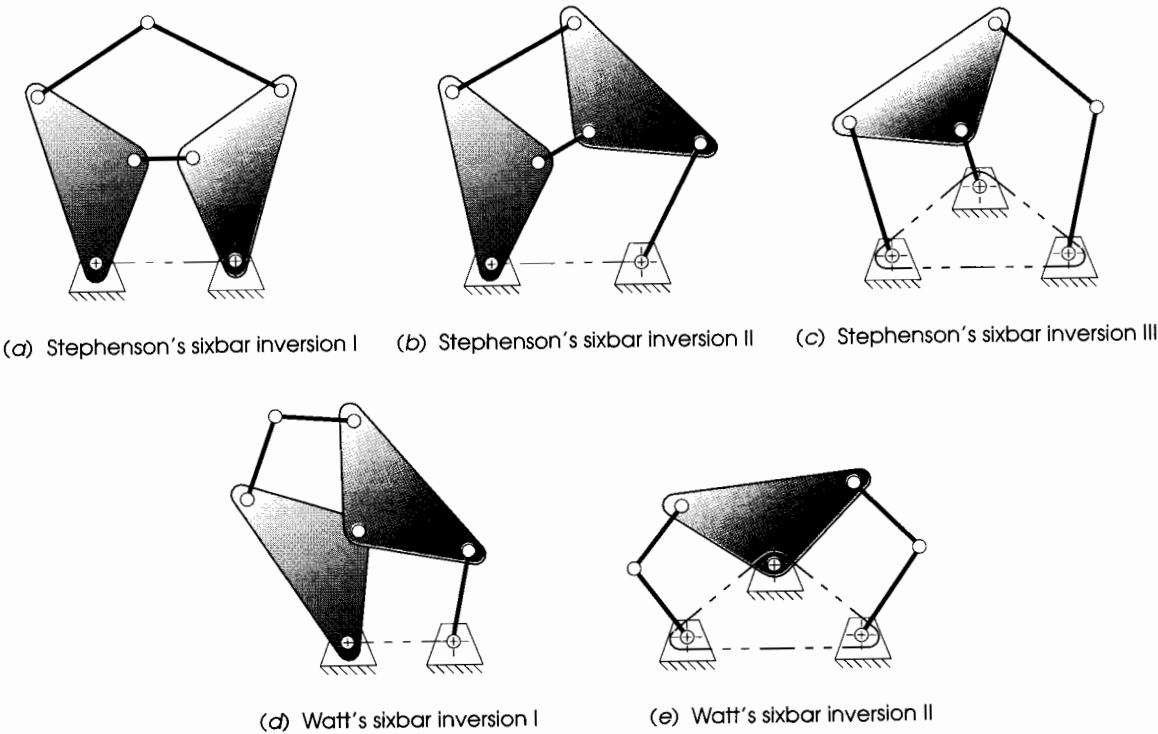
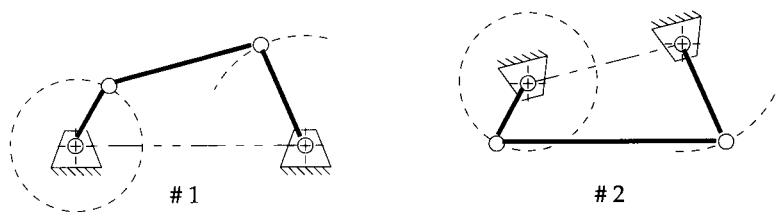
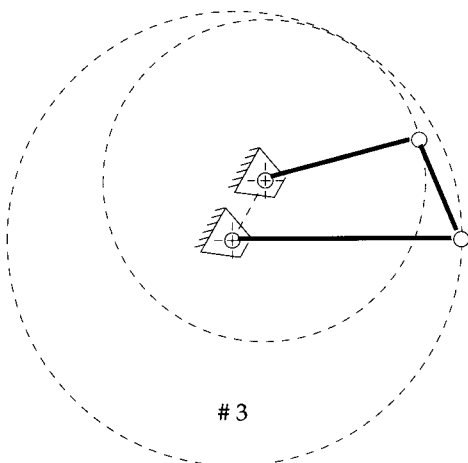
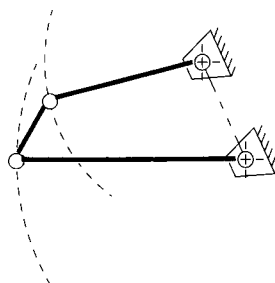


FIGURE 2-14

All distinct inversions of the sixbar linkage



(a) Two non-distinct crank-rocker inversions (GCRR)

(b) Double-crank inversion (GCRC)
(drag link mechanism)(c) Double-rocker inversion (GRCR)
(coupler rotates)**FIGURE 2-15**

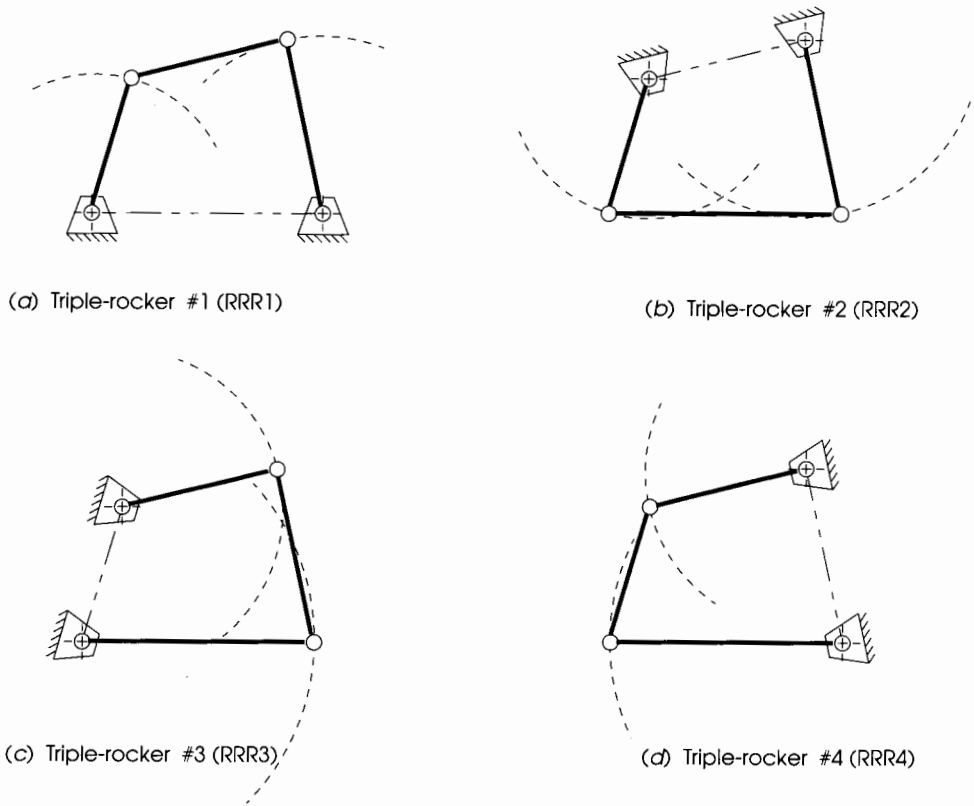
All inversions of the Grashof fourbar linkage

2.12 THE GRASHOF CONDITION

The **fourbar linkage** has been shown above to be the *simplest possible pin-jointed mechanism* for single degree of freedom controlled motion. It also appears in various disguises such as the **slider-crank** and the **earn-follower**. It is in fact the most common and ubiquitous device used in machinery. It is also extremely versatile in terms of the types of motion which it can generate.

Simplicity is one mark of good design. The fewest parts that can do the job will usually give the least expensive and most reliable solution. Thus the **fourbar linkage** should be among the first solutions to motion control problems to be investigated. The **Grashof condition** [8] is a very simple relationship which predicts the *rotation behavior* or **rotatability** of a fourbar linkage's inversions based only on the link lengths.

Let: S = length of shortest link
 L = length of longest link

**FIGURE 2-16**

All inversions of the non-Grashof fourbar linkage are triple rockers

P = length of one remaining link

Q = length of other remaining link

Then if :

$$S + L \leq P + Q \quad (2.8)$$

the linkage is Grashof and at least one link will be capable of making a full revolution with respect to the ground plane. This is called a Class I kinematic chain. If the inequality is not true, then the linkage is non-Grashof and no link will be capable of a complete revolution relative to any other link. This is a Class II kinematic chain.

Note that the above statements apply regardless of the order of assembly of the links. That is, the determination of the Grashof condition can be made on a set of unassembled links. Whether they are later assembled into a kinematic chain in S, L, P, Q , or S, P, L, Q or any other order, will *not* change the Grashof condition.

The motions possible from a fourbar linkage will depend on both the Grashof condition and the inversion chosen. The inversions will be defined with respect to the shortest link. The motions are:

For the Class I case, $S + L < P + Q$:

Ground either link adjacent to the shortest and you get a crank-rocker, in which the shortest link will fully rotate and the other link pivoted to ground will oscillate.

Ground the shortest link and you will get a double-crank, in which both links pivoted to ground make complete revolutions as does the coupler.

Ground the link opposite the shortest and you will get a Grashof double-rocker, in which both links pivoted to ground oscillate and only the coupler makes a full revolution.

For the Class II case, $S + L > P + Q$:

All inversions will be triple-rockers [9] in which no link can fully rotate.

For the Class III case, $S + L = P + Q$:

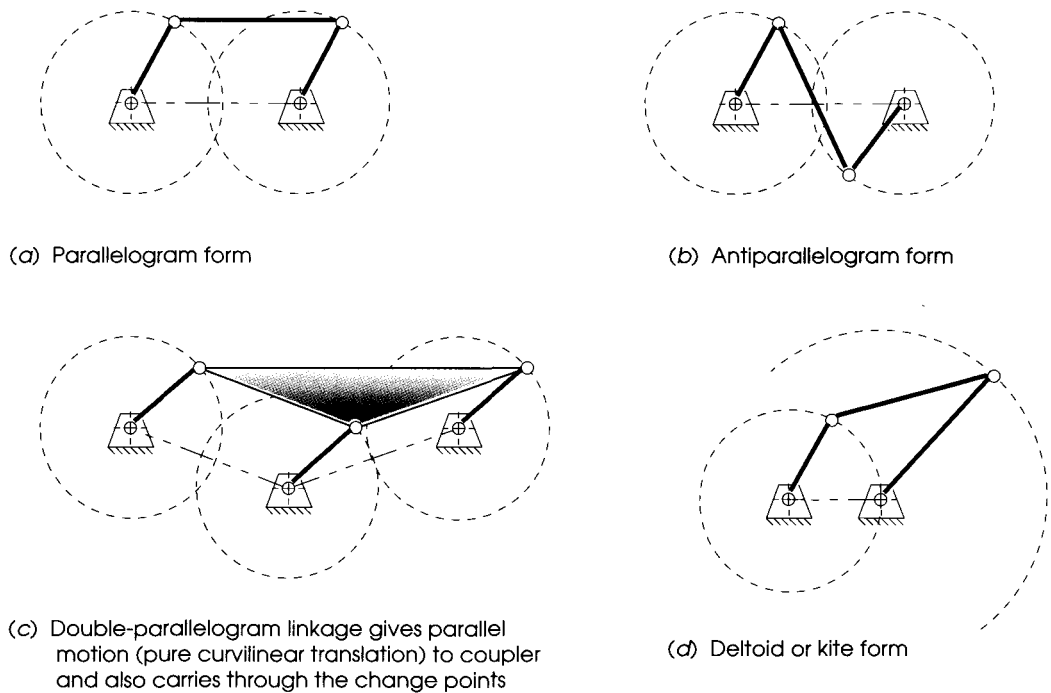
Referred to as special-case Grashof and also as a Class III kinematic chain, all inversions will be either double-cranks or crank-rockers but will have "change points" twice per revolution of the input crank when the links all become colinear. At these change points the output behavior will become indeterminate. The linkage behavior is then unpredictable as it may assume either of two configurations. Its motion must be limited to avoid reaching the change points or an additional, out-of-phase link provided to guarantee a "carry through" of the change points. (See Figure 2-17 c.)

Figure 2-15 (p. 46) shows the four possible inversions of the Grashof case: two crank-rockers, a double-crank (also called a drag link), and a double-rocker with rotating coupler. The two crank-rockers give similar motions and so are not distinct from one another. Figure 2-16 (p. 47) shows four non-distinct inversions, all triple-rockers, of a non-Grashof linkage.

Figure 2-17 a and b shows the parallelogram and antiparallelogram configurations of the special-case Grashof linkage. The parallelogram linkage is quite useful as it exactly duplicates the rotary motion of the driver crank at the driven crank. One common use is to couple the two windshield wiper output rockers across the width of the windshield on an automobile. The coupler of the parallelogram linkage is in curvilinear translation, remaining at the same angle while all points on it describe identical circular paths. It is often used for this parallel motion, as in truck tailgate lifts and industrial robots.

The antiparallelogram linkage is also a double-crank, but the output crank has an angular velocity different from the input crank. Note that the change points allow the linkage to switch unpredictably between the parallelogram and antiparallelogram forms every 180 degrees unless some additional links are provided to carry it through those positions. This can be achieved by adding an out-of-phase companion linkage coupled to the same crank, as shown in Figure 2-17c. A common application of this double parallelogram linkage was on steam locomotives, used to connect the drive wheels together. The change points were handled by providing the duplicate linkage, 90 degrees out of phase, on the other side of the locomotive's axle shaft. When one side was at a change point, the other side would drive it through.

The double-parallelogram arrangement shown in Figure 2-17 c is quite useful as it gives a translating coupler which remains horizontal in all positions. The two parallelo-

**FIGURE 2-17**

Some forms of the special-case Grashof linkage

gram stages of the linkage are out of phase so each carries the other through its change points. Figure 2-17d shows the deltoid or kite configuration which is a crank-rocker.

There is nothing either bad or good about the Grashof condition. Linkages of all three persuasions are equally useful in their place. If, for example, your need is for a motor driven windshield wiper linkage, you may want a non-special-case Grashof crank-rocker linkage in order to have a rotating link for the motor's input, plus a special-case parallelogram stage to couple the two sides together as described above. If your need is to control the wheel motions of a car over bumps, you may want a non-Grashof triple-rocker linkage for short stroke oscillatory motion. If you want to exactly duplicate some input motion at a remote location, you may want a special-case Grashof parallelogram linkage, as used in a drafting machine. In any case, this simply determined condition tells volumes about the behavior to be expected from a proposed fourbar linkage design prior to any construction of models or prototypes.

Classification of the Fourbar Linkage

Barker [10] has developed a classification scheme that allows prediction of the type of motion that can be expected from a fourbar linkage based on the values of its link ratios. A linkage's angular motion characteristics are independent of the absolute values of its link lengths. This allows the link lengths to be normalized by dividing three of them by the fourth to create three dimensionless ratios that define its geometry.

Let the link lengths be designated r_1 , r_2 , r_3 , and r_4 (all positive and nonzero), with the subscript 1 indicating the ground link, 2 the driving link, 3 the coupler, and 4 the remaining (output) link. The link ratios are then formed by dividing each link length by r_2 giving: $\lambda_1 = r_1 / r_2$, $\lambda_3 = r_3 / r_2$, $\lambda_4 = r_4 / r_2$.

Each link will also be given a letter designation based on its type of motion when connected to the other links. If a link can make a full revolution with respect to the other links, it is called a crank (C), and if not, a rocker (R). The motion of the assembled linkage based on its Grashof condition and inversion can then be given a letter code such as GCRR for a Grashof crank-rocker or GCCC for a Grashof double-crank (drag link) mechanism. The motion designators C and R are always listed in the order of, input link, coupler, output link. The prefix G indicates a Grashof linkage, S a special-case Grashof (change point), and no prefix a non-Grashof linkage.

Table 2-4 shows Barker's 14 types of fourbar linkage based on this naming scheme. The first four rows are the Grashof inversions, the next four are the non-Grashof triple rockers, and the last six are the special-case Grashof linkages. He gives unique names to each type based on a combination of their Grashof condition and inversion. The traditional names for the same inversions are also shown for comparison and are less specific than Barker's nomenclature. Note his differentiation between the Grashof crank-rocker (subclass -2) and rocker-crank (subclass -4). To drive a GRRC linkage from the

TABLE 2-4 Barker's Complete Classification of Planar Fourbar Mechanisms

Adapted from ref. (9). s = shortest link, l = longest link, Gxxx = Grashof, RRRx = non-Grashof, Sxx = Special case

Type	$s + l$ vs. $p + q$	Inversion	Class	Barker's Designation	Code	Also Known As
1	<	$L_1 = s$ = ground	I-1	Grashof crank-crank-crank	GGCC	double-crank
2	<	$L_2 = s$ = input	I-2	Grashof crank-rocker-rocker	GCRR	crank-rocker
3	<	$L_3 = s$ = coupler	I-3	Grashof rocker-crank-rocker	GRCR	double-rocker
4	<	$L_4 = s$ = output	I-4	Grashof rocker-rocker-crank	GRRC	rocker-crank
5	>	$L_1 = l$ = ground	II-1	Class 1 rocker-rocker-rocker	RRR1	triple-rocker
6	>	$L_2 = l$ = input	II-2	Class 2 rocker-rocker-rocker	RRR2	triple-rocker
7	>	$L_3 = l$ = coupler	II-3	Class 3 rocker-rocker-rocker	RRR3	triple-rocker
8	>	$L_4 = l$ = output	II-4	Class 4 rocker-rocker-rocker	RRR4	triple-rocker
9	=	$L_1 = s$ = ground	III-1	change point crank-crank-crank	SCCC	SC* double -crank
10	=	$L_2 = s$ = input	III-2	change point crank-rocker-rocker	SCRR	SC crank-rocker
11	=	$L_3 = s$ = coupler	III-3	change point rocker-crank-rocker	SRCR	SC double-rocker
12	=	$L_4 = s$ = output	III-4	change point rocker-rocker-crank	SRRC	SC rocker-crank
13	=	two equal pairs	III-5	double change point	S2X	parallelogram or deltoid
14	=	$L_1 = L_2 = L_3 = L_4$	III-6	triple change point	S3X	square

* SC = special case.

rocker requires adding a flywheel to the crank as is done with the internal combustion engine's slider-crank mechanism (which is a GPRC) linkage. See Figure 2-10a (p. 41).

Barker also defines a "solution space" whose axes are the link ratios A_i , A_3 , A_4 as shown in Figure 2-18. These ratios' values theoretically extend to infinity, but for any practical linkages the ratios can be limited to a reasonable value.

In order for the four links to be assembled, the longest link must be shorter than the sum of the other three links,

$$L < (S + P + Q) \quad (2.9)$$

If $L = (S + P + Q)$, then the links can be assembled but will not move, so this condition provides a criterion to separate regions of no mobility from regions that allow mobility within the solution space. Applying this criterion in terms of the three link ratios defines four planes of zero mobility which provide limits to the solution space.

$$\begin{aligned} 1 &= \lambda_1 + \lambda_3 + \lambda_4 \\ \lambda_3 &= \lambda_1 + 1 + \lambda_4 \\ \lambda_4 &= \lambda_1 + 1 + \lambda_3 \\ \lambda_1 &= 1 + \lambda_3 + \lambda_4 \end{aligned} \quad (2.10)$$

Applying the $S + L = P + Q$ Grashof condition (in terms of the link ratios) defines three additional planes on which the change-point mechanisms all lie.

$$\begin{aligned} 1 + \lambda_1 &= \lambda_3 + \lambda_4 \\ 1 + \lambda_3 &= \lambda_1 + \lambda_4 \\ 1 + \lambda_4 &= \lambda_1 + \lambda_3 \end{aligned} \quad (2.11)$$

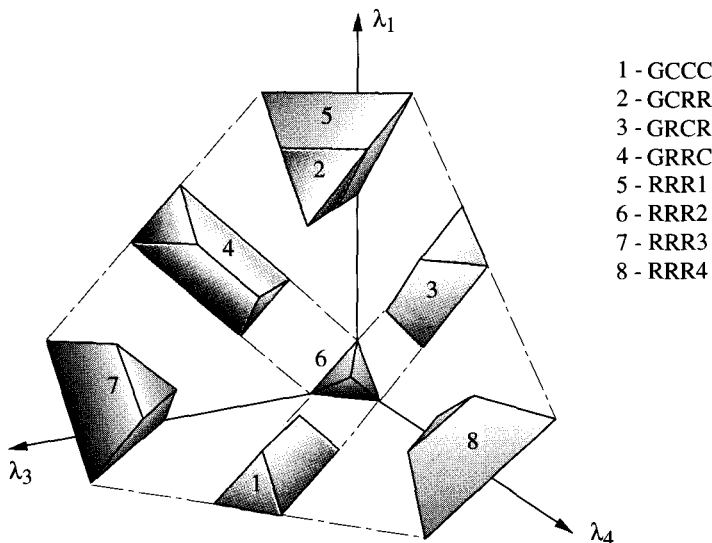


FIGURE 2-18

Barker's solution space for the fourbar linkage Adapted from reference (10).

The positive octant of this space, bounded by the $\lambda_1\text{--}\lambda_3$, $\lambda_1\text{--}\lambda_4$, $\lambda_3\text{--}\lambda_4$ planes and the four zero-mobility planes (equation 2.10) contains eight volumes that are separated by the change-point planes (equation 2.11). Each volume contains mechanisms unique to one of the first eight classifications in Table 2-4 (p. 50). These eight volumes are in contact with one another in the solution space, but to show their shapes, they have been “exploded” apart in Figure 2-18 (p. 51). The remaining six change-point mechanisms of Table 2-4 exist only in the change-point planes which are the interfaces between the eight volumes. For more detail on this solution space and Barker’s classification system than space permits here, see reference [10].

2.13 LINKAGES OF MORE THAN FOUR BARS

Gearied Fivebar Linkages

We have seen that the simplest *one-DOF* linkage is the fourbar mechanism. It is an extremely versatile and useful device. Many quite complex motion control problems can be solved with just four links and four pins. Thus in the interest of simplicity, designers should always first try to solve their problems with a fourbar linkage. However, there will be cases when a more complicated solution is necessary. Adding one link and one joint to form a fivebar (Figure 2-19a) will increase the *DOF* by one, to two. By adding a pair of gears to tie two links together with a new half joint, the *DOF* is reduced again to one, and the **gearied fivebar mechanism** (GFBM) of Figure 2-19b is created.

The geared fivebar mechanism provides more complex motions than the fourbar mechanism at the expense of the added link and gearset as can be seen in Appendix E. The reader may also observe the dynamic behavior of the linkage shown in Figure 2-19b by running the program FIVEBAR provided with this text and opening the data file F02-19b.5br. See Appendix A for instructions in running the program. Accept all the default values, and animate the linkage.

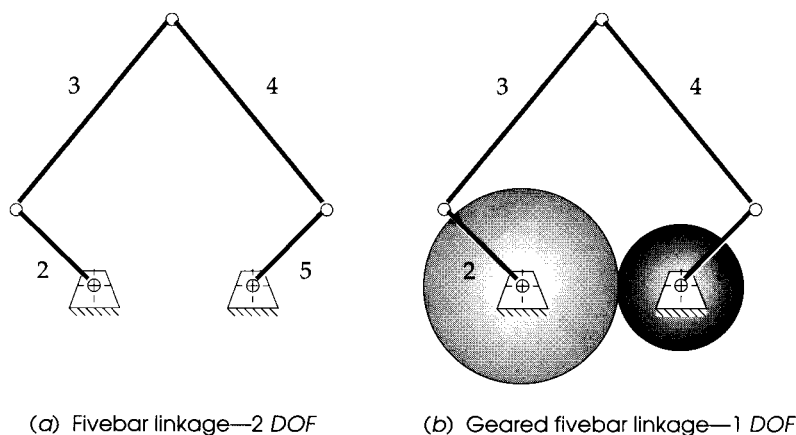


FIGURE 2-19

Two forms of the fivebar linkage

Sixbar Linkages

We already met the Watt's and Stephenson's sixbar mechanisms. See Figure 2-14 (p. 45). The Watt's sixbar can be thought of as *two fourbar linkages connected in series* and sharing two links in common. The Stephenson's sixbar can be thought of as *two fourbar linkages connected in parallel* and sharing two links in common. Many linkages can be designed by the technique of combining multiple fourbar chains as *basic building blocks* into more complex assemblages. Many real design problems will require solutions consisting of more than four bars. Some Watt's and Stephenson's linkages are provided as built-in examples to the program SIXBAR supplied with this text. You may run that program to observe these linkages dynamically. Select any example from the menu, accept all default responses, and animate the linkages.

Grashof- Type Rotatability Criteria for Higher-Order Linkages

Rotatability is defined as *the ability of at least one link in a kinematic chain to make a full revolution with respect to the other links* and defines the chain as Class I, II or III. Revolvability refers to *a specific link in a chain and indicates that it is one of the links that can rotate*.

ROTATABILITY OF GEARED FIVEBAR LINKAGES Ting [11] has derived an expression for rotatability of the geared fivebar linkage that is similar to the fourbar's Grashof criterion. Let the link lengths be designated L_1 through L_5 in order of increasing length,

$$\text{then if:} \quad L_1 + L_2 + L_5 < L_3 + L_4 \quad (2.12)$$

the two shortest links can revolve fully with respect to the others and the linkage is designated a **Class I** kinematic chain. If this inequality is *not* true, then it is a **Class II** chain and may or may not allow any links to fully rotate depending on the gear ratio and phase angle between the gears. If the inequality of equation 2.12 is replaced with an equal sign, the linkage will be a **Class III** chain in which the two shortest links can fully revolve but it will have change points like the special-case Grashof fourbar.

Reference [11] describes the conditions under which a Class II geared fivebar linkage will and will not be rotatable. In practical design terms, it makes sense to obey equation 2.12 in order to guarantee a "Grashof" condition. It also makes sense to avoid the Class III change-point condition. Note that if one of the short links (say L_2) is made zero, equation 2.12 reduces to the Grashof formula of equation 2.8.

In addition to the linkage's rotatability, we would like to know about the kinds of motions that are possible from each of the five inversions of a fivebar chain. Ting [11] describes these in detail. But, if we want to apply a gearset between two links of the fivebar chain (to reduce its *DOF* to 1), we really need it to be a double crank linkage, with the gears attached to the two cranks. A Class I fivebar chain will be a **double-crank mechanism** if *the two shortest links are among the set of three links that comprise the mechanism's ground link and the two cranks pivoted to ground*. [11]

ROTATABILITY OF N -BAR LINKAGES Ting et al. [12], [13] have extended rotatability criteria to all single-loop linkages of N -bars connected with revolute joints and have developed general theorems for **linkage rotatability** and the **revolvability** of individual links based on link lengths. Let the links of an N -bar linkage be denoted by L_i ($i = 1, 2,$

... N), with $L_1 \leq L_2 \leq \dots \leq L_N$. The links need not be connected in any particular order as rotatability criteria are independent of that factor.

A single-loop, revolute-jointed linkage of N links will have $(N - 3)$ *DOF*. The necessary and sufficient condition for the **assemblability** of an N -Bar linkage is:

$$L_N \leq \sum_{k=1}^{N-1} L_k \quad (2.13)$$

A link K will be a so-called *short* link if

$$\{K\}_{k=1}^{N-3} \quad (2.14a)$$

and a so-called *long* link if

$$\{K\}_{k=N-2}^N \quad (2.14b)$$

There will be three long links and $(N - 3)$ short links in any linkage of this type.

A single-loop N -bar kinematic chain containing only first-order revolute joints will be a Class I, Class II, or Class III linkage depending on whether the sum of the lengths of its longest link and its $(N - 3)$ shortest links is, respectively, less than, greater than, or equal to the sum of the lengths of the remaining two long links:

$$\begin{aligned} \text{Class I:} & \quad L_N + (L_1 + L_2 + \dots + L_{N-3}) < L_{N-2} + L_{N-1} \\ \text{Class II:} & \quad L_N + (L_1 + L_2 + \dots + L_{N-3}) > L_{N-2} + L_{N-1} \\ \text{Class III:} & \quad L_N + (L_1 + L_2 + \dots + L_{N-3}) = L_{N-2} + L_{N-1} \end{aligned} \quad (2.15)$$

and, for a Class I linkage, there must be one and only one long link between two non-input angles. These conditions are necessary and sufficient to define the rotatability.

The **revolvability** of any link L_i is defined as its ability to rotate fully with respect to the other links in the chain and can be determined from:

$$L_i + L_N \leq \sum_{k=1, k \neq i}^{N-1} L_k \quad (2.16)$$

Also, if L_i is a revolvable link, any link that is not longer than L_i will also be revolvable.

Additional theorems and corollaries regarding limits on link motions can be found in references [12] and [13]. Space does not permit their complete exposition here. Note that the rules regarding the behavior of geared fivebar linkages and fourbar linkages (the Grashoflaw) stated above are consistent with, and contained within, these general rotatability theorems.

2.14 SPRINGS AS LINKS

We have so far been dealing only with rigid links. In many mechanisms and machines, it is necessary to counterbalance the static loads applied to the device. A common exam-

ple is the hood hinge mechanism on your automobile. Unless you have the (cheap) model with the strut that you place in a hole to hold up the hood, it will probably have either a fourbar or sixbar linkage connecting the hood to the body on each side. The hood may be the coupler of a non-Grashof linkage whose two rockers are pivoted to the body. A spring is fitted between two of the links to provide a force to hold the hood in the open position. The spring in this case is an additional link of variable length. As long as it can provide the right amount of force, it acts to reduce the *DOF* of the mechanism to zero, and holds the system in static equilibrium. However, you can force it to again be a one-*DOF* system by overcoming the spring force when you pull the hood shut.

Another example, which may now be right next to you, is the ubiquitous adjustable arm desk lamp, shown in Figure 2-20. This device has two springs that counterbalance the weight of the links and lamp head. If well designed and made, it will remain stable over a fairly wide range of positions despite variation in the overturning moment due to the lamp head's changing moment arm. This is accomplished by careful design of the geometry of the spring-link relationships so that, as the spring force changes with increasing length, its moment arm also changes in a way that continually balances the changing moment of the lamp head.

A linear spring can be characterized by its spring constant, $k = F / x$, where F is force and x is spring displacement. Doubling its deflection will double the force. Most coil springs of the type used in these examples are linear. The design of spring-loaded linkages will be addressed in a later chapter.

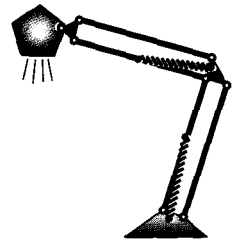


FIGURE 2-20
A spring-balanced linkage mechanism

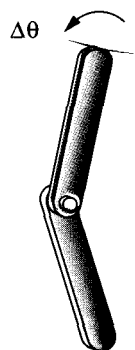
2.15 PRACTICAL CONSIDERATIONS

There are many factors that need to be considered to create good-quality designs. Not all of them are contained within the applicable theories. A great deal of art based on experience is involved in design as well. This section attempts to describe a few such practical considerations in machine design.

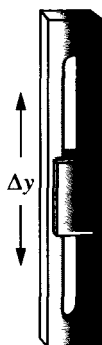
PIN JOINTS VERSUS SLIDERS AND HALF JOINTS

Proper material selection and good lubrication are the key to long life in any situation, such as a joint, where two materials rub together. Such an interface is called a bearing. Assuming the proper materials have been chosen, the choice of joint type can have a significant effect on the ability to provide good, clean lubrication over the lifetime of the machine.

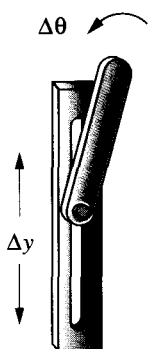
REVOLUTE (PIN) JOINTS The simple revolute or pin joint (Figure 2-21a) is the clear winner here for several reasons. It is relatively easy and inexpensive to design and build a good quality pin joint. In its pure form—a so-called sleeve or journal bearing—the geometry of pin-in-hole traps a lubricant film within its annular interface by capillary action and promotes a condition called *hydrodynamic lubrication* in which the parts are separated by a thin film of lubricant as shown in Figure 2-22. Seals can easily be provided at the ends of the hole, wrapped around the pin, to prevent loss of the lubricant. Replacement lubricant can be introduced through radial holes into the bearing interface, either continuously or periodically, without disassembly.



(a) Pin joint



(b) Slider joint



(c) Half joint

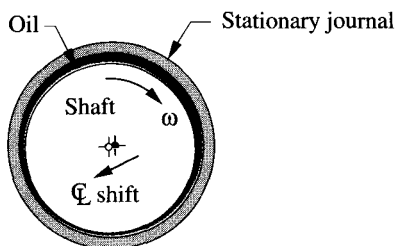
A convenient form of bearing for linkage pivots is the commercially available spherical rod end shown in Figure 2-23. This has a spherical, sleeve-type bearing which *self-aligns* to a shaft that may be out of parallel. Its body threads onto the link, allowing links to be conveniently made from round stock with threaded ends that allow adjustment of link length.

Relatively inexpensive ball and roller bearings are commercially available in a large variety of sizes for revolute joints as shown in Figure 2-24. Some of these bearings (principally ball type) can be obtained prelubricated and with end seals. Their rolling elements provide low-friction operation and good dimensional control. Note that *rolling-element bearings* actually contain higher-joint interfaces (half joints) at each ball or roller, which is potentially a problem as noted below. However, the ability to trap lubricant within the roll cage (by end seals) combined with the relatively high rolling speed of the balls or rollers promotes hydrodynamic lubrication and long life. For more detailed information on bearings and lubrication, see reference [15].

For revolute joints pivoted to ground, several commercially available bearing types make the packaging easier. Pillow blocks and flange-mount bearings (Figure 2-25) are available fitted with either rolling-element (ball, roller) bearings or sleeve-type journal bearings. The pillow block allows convenient mounting to a surface parallel to the pin axis, and flange mounts fasten to surfaces perpendicular to the pin axis.

PRISMATIC (SLIDER) JOINTS require a carefully machined and straight slot or rod (Figure 2-21b). The bearings often must be custom made, though linear ball bearings (Figure 2-26) are commercially available but must be run over hardened and ground shafts. Lubrication is difficult to maintain in any sliding joint. The lubricant is not geometrically captured, and it must be resupplied either by running the joint in an oil bath or by periodic manual regreasing. An open slot or shaft tends to accumulate airborne dirt particles which can act as a grinding compound when trapped in the lubricant. This will accelerate wear.

HIGHER (HALF) JOINTS such as a round pin in a slot (Figure 2-21c) or a cam-follower joint (Figure 2-10c, p. 41) suffer even more acutely from the slider's lubrication problems, because they typically have two oppositely curved surfaces in line contact, which tend to squeeze any lubricant out of the joint. This type of joint needs to be run in an oil bath for long life. This requires that the assembly be housed in an expensive, oil-tight box with seals on all protruding shafts.



- Shaft rotating rapidly
- hydrodynamic lubrication
 - no metal contact
 - fluid pumped by shaft
 - shaft lags bearing centerline

FIGURE 2-21

Joints of various types

FIGURE 2-22

Hydrodynamic lubrication in a sleeve bearing—clearance and motions exaggerated

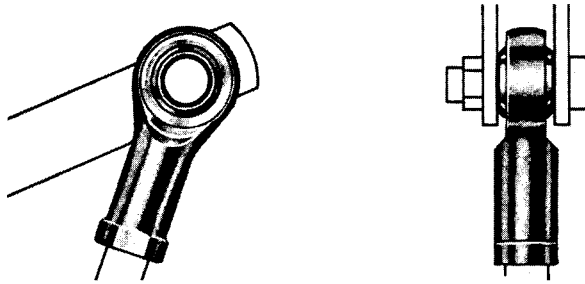
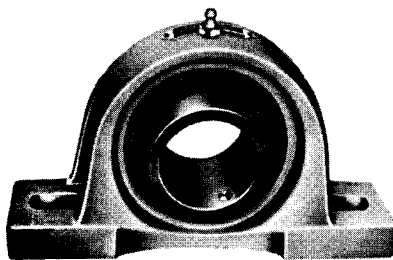


FIGURE 2-23
Spherical rod end *Courtesy of Emerson Power Transmission, Ithaca, NY.*

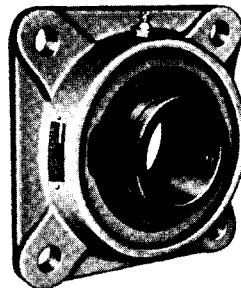
These joint types are all used extensively in machinery with great success. As long as the proper *attention to engineering detail* is paid, the design can be successful. Some common examples of all three joint types can be found in an automobile. The windshield wiper mechanism is a pure pin-jointed linkage. The pistons in the engine cylinders are true sliders and are bathed in engine oil. The valves in the engine are opened and closed by earn-follower (halt) joints which are drowned in engine oil. You probably change your engine oil fairly frequently. When was the last time you lubricated your windshield wiper linkage? Has this linkage (not the motor) ever failed?

Cantilever or Straddle Mount?

Any joint must be supported against the joint loads. Two basic approaches are possible as shown in Figure 2-27. A cantilevered joint has the pin Goumal) supported only, as a cantilever beam. This is sometimes necessary as with a crank that must pass over the coupler and cannot have anything on the other side of the coupler. However, a cantile

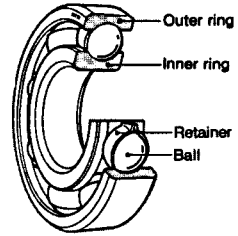


(a) Pillow-block bearing

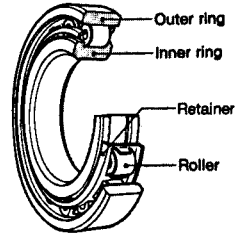


(b) Flange-mount bearing

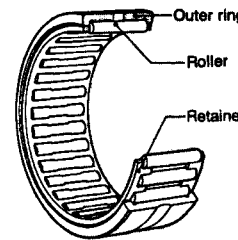
FIGURE 2-25
Pillow block and flange-mount bearing units. *Courtesy of Emerson Power Transmission, Ithaca, NY.*



(a) Ball bearing



(b) Roller bearing



(c) Needle bearing

FIGURE 2-24
Ball, roller and needle bearings for revolute joints. *Courtesy of NTN Corporation, Japan.*

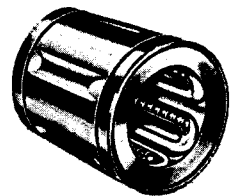
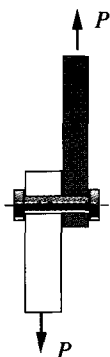
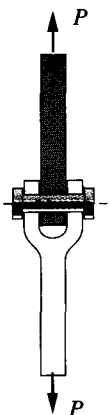


FIGURE 2-26
Linear ball bushing
Courtesy of Thomson Industries, Port Washington, NY



(a) Cantilever mount
—single shear



(b) Straddle mount
—double shear

FIGURE 2-27

Cantilever, and
straddle-mounted pin
joints



FIGURE 2-28

Shoulder screw
Courtesy of Cordova Bolt
Inc., Buena Park, CA

ver beam is inherently weaker (for the same cross section and load) than a straddle-mounted (simply supported) beam. The straddle mounting can avoid applying a bending moment to the links by keeping the forces in the same plane. The pin will feel a bending moment in both cases, but the straddle-mounted pin is in double shear—two cross sections are sharing the load. A cantilevered pin is in single shear. It is good practice to use straddle-mounted joints (whether revolute, prismatic, or higher) wherever possible. If a cantilevered pin must be used, then a commercial shoulder screw that has a hardened and ground shank as shown in Figure 2-28 can sometimes serve as a pivot pin.

Short Links

It sometimes happens that the required length of a crank is so short that it is not possible to provide suitably sized pins or bearings at each of its pivots. The solution is to design the link as an **eccentric crank**, as shown in Figure 2-29. One pivot pin is enlarged to the point that it, in effect, contains the link. The outside diameter of the circular crank becomes the journal for the moving pivot. The fixed pivot is placed a distance e from the center of this circle equal to the required crank length. The distance e is the crank's eccentricity (the crank length). This arrangement has the advantage of a large surface area within the bearing to reduce wear, though keeping the large-diameter journal lubricated can be difficult.

Bearing Ratio

The need for straight-line motion in machinery requires extensive use of linear translating slider joints. There is a very basic geometrical relationship called *bearing ratio*, which if ignored or violated will invariably lead to problems.

The **bearing ratio (BR)** is defined as *the effective length of the slider over the effective diameter of the bearing*: $BR = L / D$. For smooth operation **this ratio should be greater than 1.5, and never less than 1**. The larger it is, the better. **Effective length** is defined as *the distance over which the moving slider contacts the stationary guide*. There need not be continuous contact over that distance. That is, two short collars, spaced far apart, are effectively as long as their overall separation plus their own lengths and are kinematically equivalent to a long tube. **Effective diameter** is *the largest distance across the stationary guides*, in any plane perpendicular to the sliding motion.

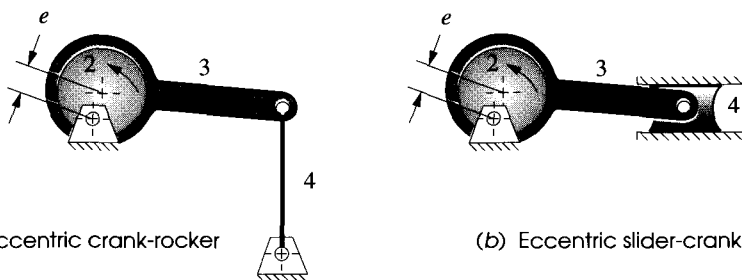
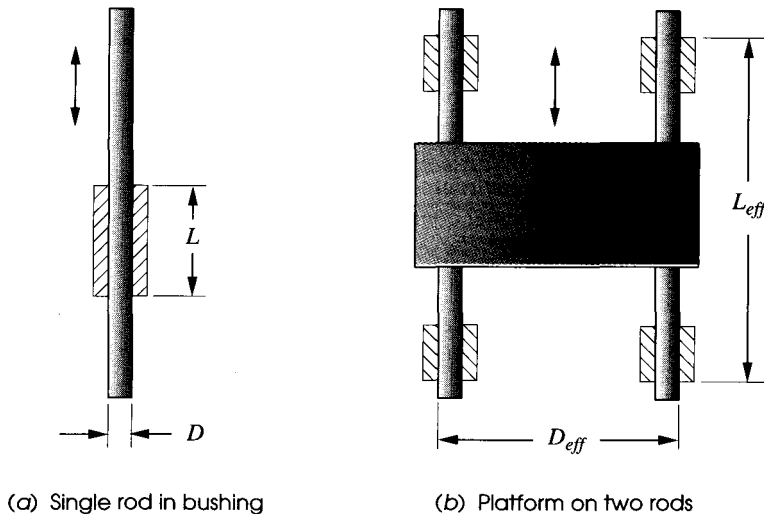


FIGURE 2-29

Eccentric cranks

**FIGURE 2-30**

Bearing ratio

If the slider joint is simply a rod in a bushing, as shown in Figure 2-30a, the effective diameter and length are identical to the actual dimensions of the rod diameter and bushing length. If the slider is a platform riding on two rods and multiple bushings, as shown in Figure 2-30b, then the effective diameter and length are the overall width and length, respectively, of the platform assembly. It is this case that often leads to poor bearing ratios.

A common example of a device with a poor bearing ratio is a drawer in an inexpensive piece of furniture. If the only guides for the drawer's sliding motion are its sides running against the frame, it will have a bearing ratio less than 1, since it is wider than it is deep. You have probably experienced the sticking and jamming that occurs with such a drawer. A better-quality chest of drawers will have a center guide with a large L/D ratio under the bottom of the drawer and will slide smoothly.

Unkages versus Cams

The pin-jointed linkage has all the advantages of revolute joints listed above. The earn-follower mechanism (Figure 2-10c, p. 41) has all the problems associated with the half joint listed above. But, both are widely used in machine design, often in the same machine and in combination (cams driving linkages). So why choose one over the other?

The "pure" pin-jointed linkage with good bearings at the joints is a potentially superior design, all else equal, and it should be the first possibility to be explored in any machine design problem. However, there will be many problems in which the need for a straight, sliding motion or the exact dwells of a earn-follower are required. Then the practical limitations of earn and slider joints will have to be dealt with accordingly.

Linkages have the disadvantage of relatively large size compared to the output displacement of the working portion; thus they can be somewhat difficult to package. Cams tend to be compact in size compared to the follower displacement. Linkages are relatively difficult to synthesize, and cams are relatively easy to design (as long as a computer is available). But linkages are much easier and cheaper to manufacture to high precision than cams. Dwells are easy to get with cams, and difficult with linkages. Linkages can survive very hostile environments, with poor lubrication, whereas cams cannot, unless sealed from environmental contaminants. Linkages have better high-speed dynamic behavior than cams, are less sensitive to manufacturing errors, and can handle very high loads, but cams can match specified motions better.

So the answer is far from clear-cut. It is another *design trade-off situation* in which you must weigh all the factors and make the best compromise. Because of the potential advantages of the pure linkage it is important to consider a linkage design before choosing a potentially easier design task but an ultimately more expensive solution.

2.16 MOTORS AND DRIVERS

Unless manually operated, a mechanism will require some type of driver device to provide the input motion and energy. There are many possibilities. If the design requires a continuous rotary input motion, such as for a Grashof linkage, a slider-crank, or a cam-follower, then a motor or engine* is the logical choice. Motors come in a wide variety of types. The most common energy source for a motor is electricity, but compressed air and pressurized hydraulic fluid are also used to power air and hydraulic motors. Gasoline or diesel engines are another possibility. If the input motion is translation, as is common in earth-moving equipment, then a hydraulic or pneumatic cylinder is usually needed.

Electric Motors

Electric motors are classified both by their function or application and by their electrical configuration. Some functional classifications (described below) are gearmotors, servomotors, and stepping motors. Many different electrical configurations as shown in Figure 2-31 are also available, independent of their functional classifications. The main electrical configuration division is between AC and DC motors, though one type, the universal motor is designed to run on either AC or DC.

AC and DC refer to *alternating current* and *direct current* respectively. AC is typically supplied by the power companies and, in the U. S., will be alternating sinusoidally at 60 hertz (Hz), at about ± 120 , ± 240 , or ± 480 volts (V) peak. Many other countries supply AC at 50 Hz. Single-phase AC provides a single sinusoid varying with time, and 3-phase AC provides three sinusoids at 120° phase angles. DC current is constant with time, supplied from generators or battery sources and is most often used in vehicles, such as ships, automobiles, aircraft, etc. Batteries are made in multiples of 1.5 V, with 6, 12, and 24 V being the most common. Electric motors are also classed by their rated power as shown in Table 2-5. Both AC and DC motors are designed to provide continuous rotary output. While they can be stalled momentarily against a load, they can not tolerate a full-current, zero-velocity stall for more than a few minutes without overheating.

* The terms motor and engine are often used interchangeably, but they do not mean the same thing. Their difference is largely semantic, but the "purist" reserves the term *motor* for electrical, hydraulic and pneumatic motors and the term *engine* for thermodynamic devices such as steam engines and internal combustion engines. Thus, your automobile is powered by an engine, but its windshield wipers and window lifts are run by motors.

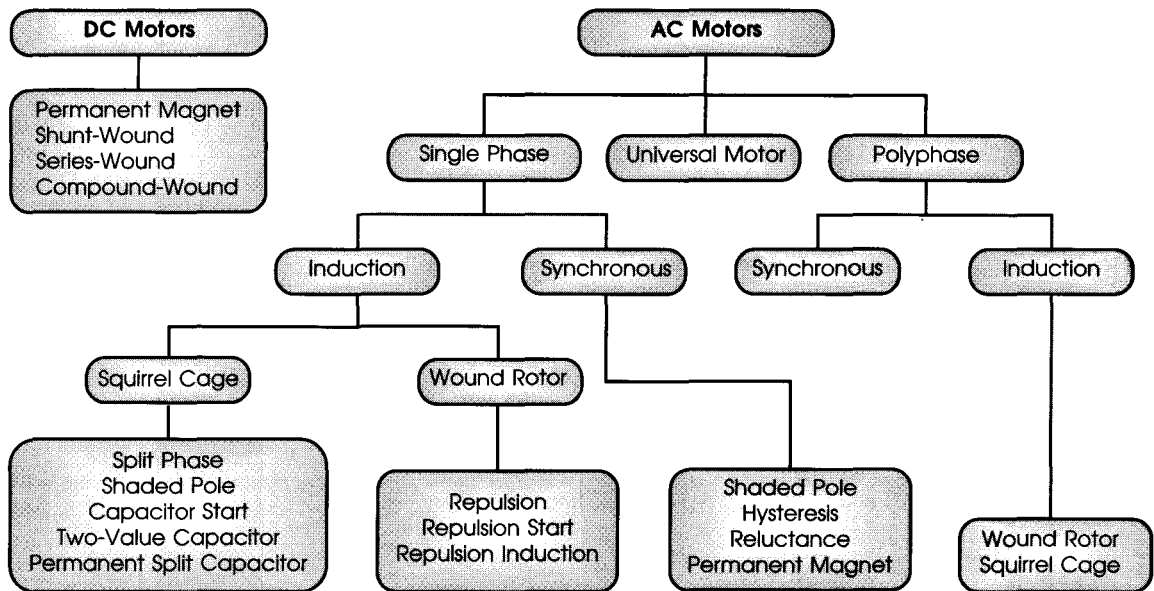


FIGURE 2-31
Types of electric motors Source: Reference (14)

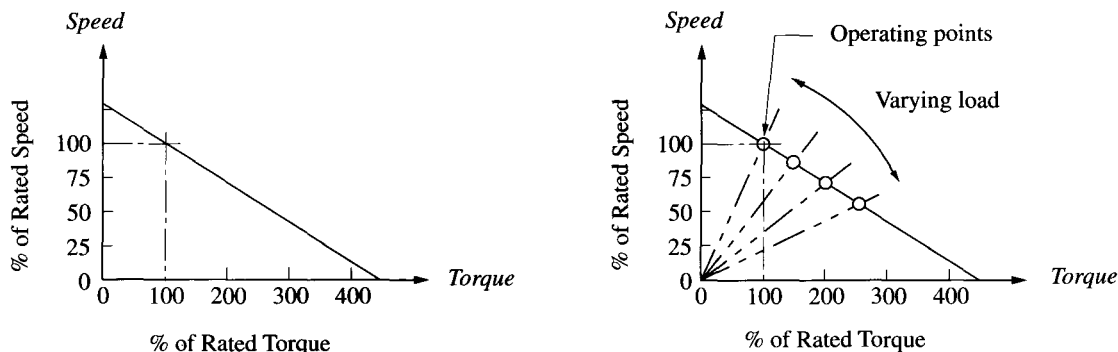
DC MOTORS are made in different electrical configurations, such as *permanent magnet (PM)*, *shunt-wound*, *series-wound*, and *compound-wound*. The names refer to the manner in which the rotating armature coils are electrically connected to the stationary field coils—in parallel (shunt), in series, or in combined series-parallel (compound). Permanent magnets replace the field coils in a PM motor. Each configuration provides different *torque-speed* characteristics. The *torque-speed* curve of a motor describes how it will respond to an applied load and is of great interest to the mechanical designer as it predicts how the mechanical-electrical system will behave when the load varies dynamically with time.

PERMANENT MAGNET DC MOTORS Figure 2-32a shows a torque-speed curve for a permanent magnet (PM) motor. Note that its torque varies greatly with speed, ranging from a maximum (stall) torque at zero speed to zero torque at maximum (no-load) speed. This relationship comes from the fact that $power = torque \times angular\ velocity$. Since the power available from the motor is limited to some finite value, an increase in torque requires a decrease in angular velocity and vice versa. Its torque is maximum at stall (zero velocity), which is typical of all electric motors. This is an advantage when starting heavy loads: e.g., an electric-motor-powered vehicle needs no clutch, unlike one powered by an internal combustion engine which cannot start from stall under load. An engine's torque increases rather than decreases with increasing angular velocity.

Figure 2-32b shows a family of **load lines** superposed on the *torque-speed* curve of a PM motor. These load lines represent a time-varying load applied to the driven mechanism. The problem comes from the fact that *as the required load torque increases, the motor must reduce speed to supply it*. Thus, the input speed will vary in response to load

TABLE 2-5
Motor Power Classes

Class	HP
Subfractional	< 1/20
Fractional	1/20 – 1
Integral	> 1



(a) Speed-torque characteristic of a PM electric motor (b) Load lines superposed on speed-torque curve

FIGURE 2-32

DC permanent magnet (PM) electric motor's typical speed-torque characteristic

variations in most motors, regardless of their design. * If constant speed is required, this may be unacceptable. Other types of DC motors have either more or less speed sensitivity to load than the PM motor. *A motor is typically selected based on its torque-speed curve.*

SHUNT-WOUND DC MOTORS have a torque speed curve like that shown in Figure 2-33a. Note the flatter slope around the rated torque point (at 100%) compared to Figure 2-32. The shunt-wound motor is less speed-sensitive to load variation in its operating range, but stalls very quickly when the load exceeds its maximum overload capacity of about 250% of rated torque. Shunt-wound motors are typically used on fans and blowers.

SERIES-WOUND DC MOTORS have a torque-speed characteristic like that shown in Figure 2-33b. This type is more speed-sensitive than the shunt or PM configurations. However, its starting torque can be as high as 800% of full-load rated torque. It also does not have any theoretical maximum no-load speed which makes it tend to run away if the load is removed. Actually, friction and windage losses will limit its maximum speed which can be as high as 20,000 to 30,000 revolutions per minute (rpm). Overspeed detectors are sometimes fitted to limit its unloaded speed. Series-wound motors are used in sewing machines and portable electric drills where their speed variability can be an advantage as it can be controlled, to a degree, with voltage variation. They are also used in heavy-duty applications such as vehicle traction drives where their high starting torque is an advantage. Also their speed sensitivity (large slope) is advantageous in high-load applications as it gives a "soft-start" when moving high-inertia loads. The motor's tendency to slow down when the load is applied cushions the shock that would be felt if a large step in torque were suddenly applied to the mechanical elements.

* The synchronous AC motor and the speed-controlled DC motor are exceptions.

COMPOUND-WOUND DC MOTORS have their field and armature coils connected in a combination of series and parallel. As a result their torque-speed characteristic has aspects of both the shunt-wound and series-wound motors as shown in Figure 2-33c.

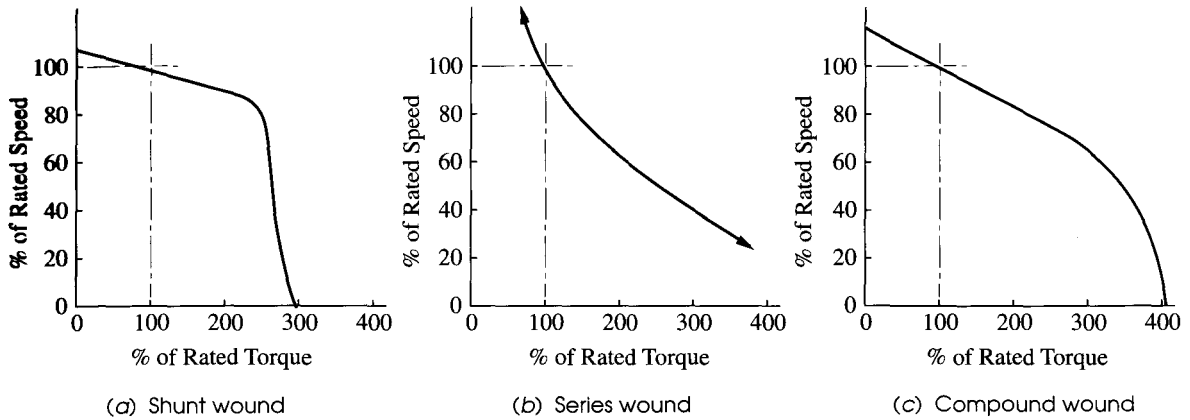


FIGURE 2-33

Torque-speed curves for three types of DC motor

their speed sensitivity is greater than a shunt-wound but less than a series-wound motor and it will not run away when unloaded. This feature plus its high starting torque and soft-start capability make it a good choice for cranes and hoists which experience high inertial loads and can suddenly lose the load due to cable failure, creating a potential run-away problem if the motor does not have a self-limited no-load speed.

SPEED-CONTROLLED DC MOTORS If precise speed control is needed, as is often the case in production machinery, another solution is to use a speed-controlled DC motor which operates from a controller that increases and decreases the current to the motor in the face of changing load to try to maintain constant speed. These speed-controlled (typically PM) DC motors will run from an AC source since the controller also converts AC to DC. The cost of this solution is high, however. Another possible solution is to provide a flywheel on the input shaft, which will store kinetic energy and help smooth out the speed variations introduced by load variations. Flywheels will be investigated in Chapter 11.

AC MOTORS are the least expensive way to get continuous rotary motion, and they can be had with a variety of *torque-speed* curves to suit various load applications. They are limited to a few standard speeds that are a function of the AC line frequency (60 Hz in North America, 50 Hz elsewhere). The synchronous motor speed n_s is a function of line frequency f and the number of magnetic poles p present in the rotor.

$$n_s = \frac{120f}{p} \quad (2.17)$$

Synchronous motors "lock on" to the AC line frequency and run exactly at synchronous speed. These motors are used for clocks and timers. Nonsynchronous AC motors have a small amount of slip which makes them lag the line frequency by about 3 to 10%.

Table 2-6 shows the synchronous and non synchronous speeds for various AC motor-pole configurations. The most common AC motors have 4 poles, giving nonsynchro-

TABLE 2-6
AC Motor Speeds

Poles	Sync rpm	Async rpm
2	3600	3450
4	1800	1725
6	1200	1140
8	900	850
10	720	690
12	600	575

nous *no-load speeds* of about 1725 rpm, which reflects slippage from the 60-Hz synchronous speed of 1800 rpm.

Figure 2-34 shows typical torque-speed curves for single-phase (1 ϕ) and 3-phase (3 ϕ) AC motors of various designs. The single-phase shaded pole and permanent split capacitor designs have a starting torque lower than their full-load torque. To boost the start torque, the split-phase and capacitor-start designs employ a separate starting circuit that is cut off by a centrifugal switch as the motor approaches operating speed. The broken curves indicate that the motor has switched from its starting circuit to its running circuit. The NEMA * three-phase motor designs B, C, and D in Figure 2-34 differ mainly in their starting torque and in speed sensitivity (slope) near the full-load point.

GEARMOTORS If different single (as opposed to variable) output speeds than the standard ones of Table 2-6 are needed, a gearbox speed reducer can be attached to the motor's output shaft, or a gearmotor can be purchased that has an integral gearbox. Gearmotors are commercially available in a large variety of output speeds and power ratings. The kinematics of gearbox design are covered in Chapter 9.

SERVOMOTORS are fast-response, closed-loop-controlled motors capable of providing a programmed function of acceleration or velocity, as well as of holding a fixed position against a load. **Closed loop** means that *sensors on the output device being moved feed back information on its position, velocity, and acceleration*. Circuitry in the motor controller responds to the fed back information by reducing or increasing (or reversing) the current flow to the motor. Precise positioning of the output device is then possible, as is control of the speed and shape of the motor's response to changes in load or input commands. These are very expensive devices which are commonly used in applications such as moving the flight control surfaces in aircraft and guided missiles, and in controlling robots, for example. Servomotors have lower power and torque capacity than is available from non servo AC or DC motors.

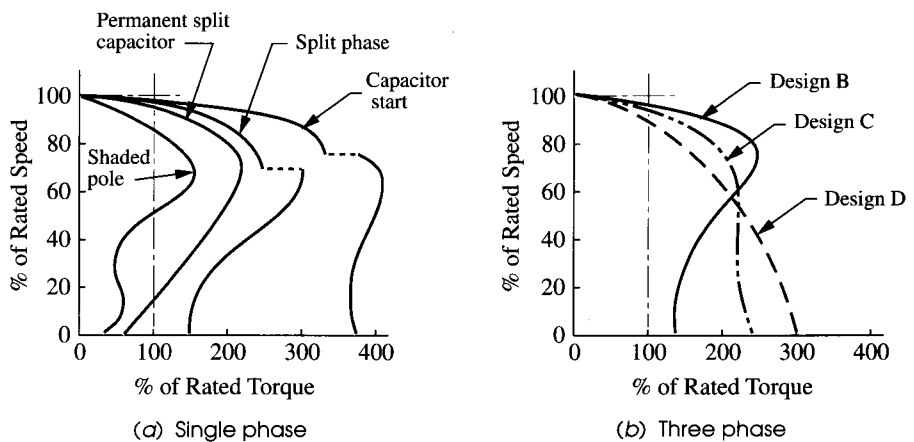


FIGURE 2-34

Torque-speed curves for single- and three-phase AC motors

* National Electrical
Manufacturers Association.

STEPPER MOTORS are designed to position an output device. Unlike servomotors, these run **open loop**, meaning they *receive no feedback as to whether the output device has responded as requested*. Thus they can get out of phase with the desired program. They will, however, happily sit energized for an indefinite period, holding the output in one position. Their internal construction consists of a number of magnetic strips arranged around the circumference of both the rotor and stator. When energized, the rotor will move one step, to the next magnet, for each pulse received. Thus, these are **intermittent motion** devices and do not provide continuous rotary motion like other motors. The number of magnetic strips determines their resolution (typically a few degrees per step). They are relatively small compared to AC/DC motors and have low torque capacity. They are moderately expensive and require special controllers.

Air and Hydraulic Motors

These have more limited application than electric motors, simply because they require the availability of a compressed air or hydraulic source. Both of these devices are less energy efficient than the direct electrical to mechanical conversion of electric motors, because of the losses associated with the conversion of the energy first from chemical or electrical to fluid pressure and then to mechanical form. Every energy conversion involves some losses. Air motors find widest application in factories and shops, where high-pressure compressed air is available for other reasons. A common example is the air impact wrench used in automotive repair shops. Although individual air motors and air cylinders are relatively inexpensive, these pneumatic systems are quite expensive when the cost of all the ancillary equipment is included. Hydraulic motors are most often found within machines or systems such as construction equipment (cranes), aircraft, and ships, where high-pressure hydraulic fluid is provided for many purposes. Hydraulic systems are very expensive when the cost of all the ancillary equipment is included.

Air and Hydraulic Cylinders

These are linear actuators (piston in cylinder) which provide a limited stroke, straight-line output from a pressurized fluid flow input of either compressed air or hydraulic fluid (usually oil). They are the method of choice if you need a linear motion as the input. However, they share the same high cost, low efficiency, and complication factors as listed under their air and hydraulic motor equivalents above.

Another problem is that of control. Most motors, left to their own devices, will tend to run at a constant speed. A linear actuator, when subjected to a constant pressure fluid source, typical of most compressors, will respond with more nearly constant acceleration, which means its velocity will increase linearly with time. This can result in severe impact loads on the driven mechanism when the actuator comes to the end of its stroke at maximum velocity. Servovalve control of the fluid flow, to slow the actuator at the end of its stroke, is possible but is quite expensive.

The most common application of fluid power cylinders is in farm and construction equipment such as tractors and bulldozers, where open loop (non servo) hydraulic cylinders actuate the bucket or blade through linkages. The cylinder and its piston become two of the links (slider and track) in a slider-crank mechanism. See Figure I-1b (p. 7).

Solenoids

These are electromechanical (AC or DC) linear actuators which share some of the limitations of air cylinders, and they possess a few more of their own. They are *energy inefficient*, are limited to very short strokes (about 2 to 3 cm), develop a force which varies exponentially over the stroke, and deliver high impact loads. They are, however, inexpensive, reliable, and have very rapid response times. They cannot handle much power, and they are typically used as control or switching devices rather than as devices which do large amounts of work on a system.

A common application of solenoids is in camera shutters, where a small solenoid is used to pull the latch and trip the shutter action when you push the button to take the picture. Its nearly instantaneous response is an asset in this application, and very little work is being done in tripping a latch. Another application is in electric door or trunk locking systems in automobiles, where the click of their impact can be clearly heard when you turn the key (or press the button) to lock or unlock the mechanism.

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2.18 PROBLEMS

- *2-1 Find three (or other number as assigned) of the following common devices. Sketch careful kinematic diagrams and find their total degrees of freedom.
- An automobile hood hinge mechanism
 - An automobile hatchback lift mechanism
 - An electric can opener
 - A folding ironing board
 - A folding card table
 - A folding beach chair
 - A baby swing
 - A folding baby walker
 - A drafting machine
 - A fancy corkscrew
 - A windshield wiper mechanism
 - A dump truck dumpster mechanism
 - A trash truck dumpster mechanism
 - A station wagon tailgate mechanism
 - An automobile jack
 - A collapsible auto radio antenna
 - A record turntable and tone arm
- 2-2 How many *DOF* do you have in your wrist and hand combined?
- 2-3 How many *DOF* do the following joints have?
- Your knee
 - Your ankle
 - Your shoulder
 - Your hip
 - Your knuckle
- 2-4 How many *DOF* do the following have in their normal environment?
- A submerged submarine
 - An earth-orbiting satellite
 - A surface ship
 - A motorcycle
 - The print head in a 9-pin dot matrix computer printer
 - The pen in an XY plotter
- 2-5 Are the joints in Problem 2-3 force closed or form closed?
- 2-6 Describe the motion of the following items as pure rotation, pure translation, or complex planar motion.
- A windmill
 - A bicycle (in the vertical plane, not turning)
 - A conventional "double-hung" window
 - The keys on a computer keyboard
 - The hand of a clock
 - A hockey puck on the ice

- g. The pen in an XY plotter
 - h. The print head in a computer printer
 - i. A "casement" window
- *2-7 Calculate the *DOF* of the linkages shown in Figure P2-1.
- *2-8 Identify the items in Figure P2-1 as mechanisms, structures, or preloaded structures.
- *2-9 Use linkage transformation on the linkage of Figure P2-1a to make it a *1-DOF* mechanism.
- *2-10 Use linkage transformation on the linkage of Figure P2-1d to make it a *2-DOF* mechanism.
- 2-11 Use number synthesis to find all the possible link combinations for *2-DOF*, up to 9 links, to hexagonal order, using only revolute joints.
- 2-12 Find all the valid isomers of the eightbar *1-DOF* link combinations in Table 2-2 (p. 36) having:
- a. Four binary and four ternary links
 - b. Five binaries, two ternaries, and one quaternary link
 - c. Six binaries and two quaternary links
 - d. Six binaries, one ternary, and one pentagonal link
- 2-13 Use linkage transformation to create a *1-DOF* mechanism with two sliding full joints from a Stephenson's sixbar linkage in Figure 2-14a (p. 45).

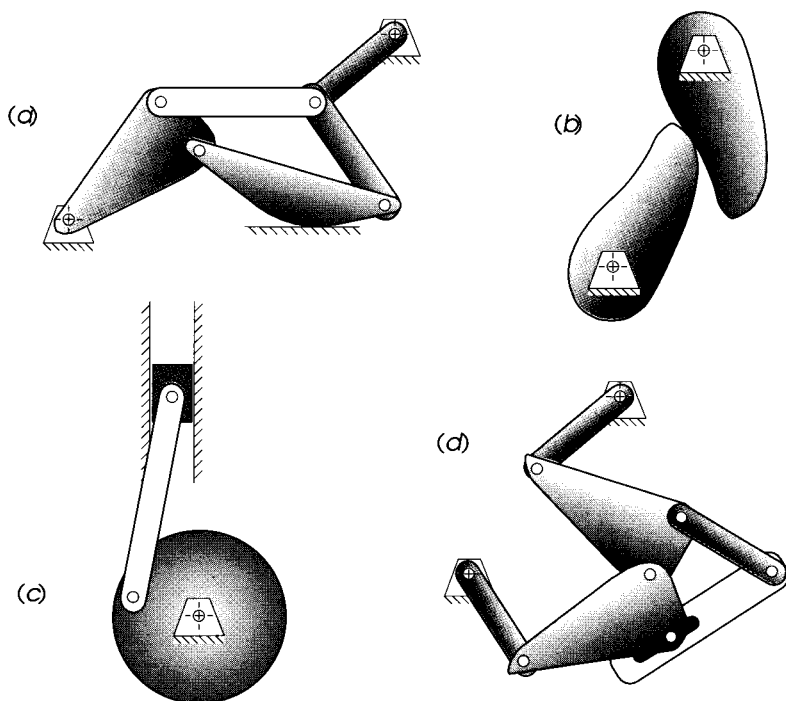


FIGURE P2-1

Linkages for Problems 2-7 to 2-10

- 2-14 Use linkage transformation to create a *I-DOF* mechanism with one sliding full joint and a half joint from a Stephenson's sixbar linkage in Figure 2-14b.
- *2-15 Calculate the Grashof condition of the fourbar mechanisms defined below. Build cardboard models of the linkages and describe the motions of each inversion. Link lengths are in inches (or double given numbers for centimeters).
- | | | | | |
|----|---|-----|---|---|
| a. | 2 | 4.5 | 7 | 9 |
| b. | 2 | 3.5 | 7 | 9 |
| c. | 2 | 4.0 | 6 | 8 |
- 2-16 What type(s) of electric motor would you specify
- To drive a load with large inertia.
 - To minimize variation of speed with load variation.
 - To maintain accurate constant speed regardless of load variations.
- 2-17 Describe the difference between a cam-follower (half) joint and a pin joint.
- 2-18 Examine an automobile hood hinge mechanism of the type described in Section 2.14. Sketch it carefully. Calculate its *DOF* and Grashof condition. Make a cardboard model. Analyze it with a free-body diagram. Describe how it keeps the hood up.
- 2-19 Find an adjustable arm desk lamp of the type shown in Figure P2-2. Measure it and sketch it to scale. Calculate its *DOF* and Grashof condition. Make a cardboard model. Analyze it with a free-body diagram. Describe how it keeps itself stable. Are there any positions in which it loses stability? Why?
- 2-20 Make kinematic sketches, define the types of all the links and joints, and determine the *DOF* of the mechanisms shown in Figure P2-3.
- *2-21 Find the *DOF* of the mechanisms in Figure P2-4.
- 2-22 Find the Grashof condition and Barker classifications of the mechanisms in Figure P2-4a, b, and d. Scale the diagrams for dimensions.

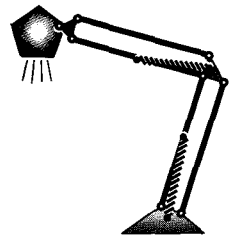


FIGURE P2-2
Problem 2-19

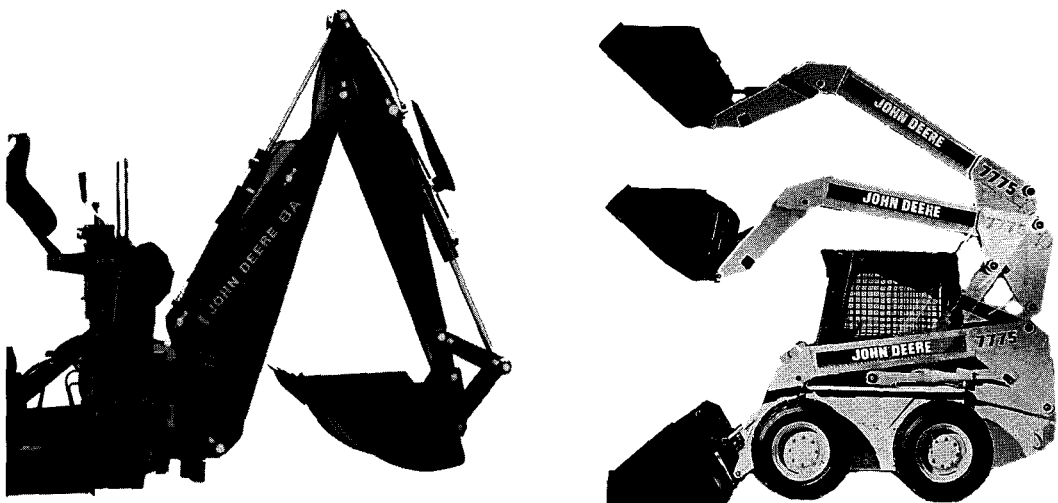


FIGURE P2-3

Problem 2-20 Backhoe and front-end loader. Courtesy of John Deere Co.

* Answers in Appendix F

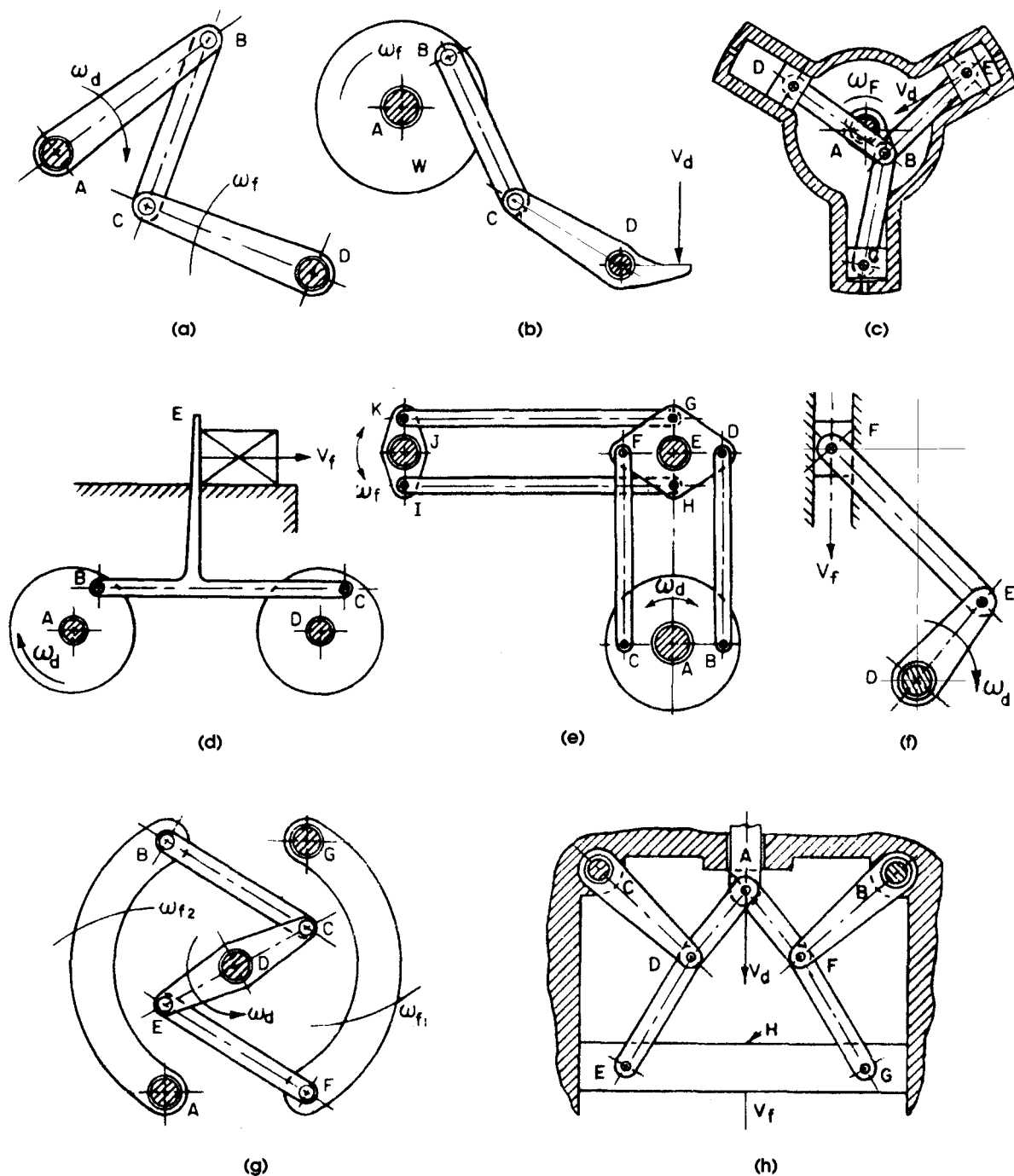


FIGURE P2-4

Problems 2-21 to 2-23 Adapted from P. H. Hill and W. P. Rule. (1960) *Mechanisms: Analysis and Design*, with permission



FIGURE P2-5

Problem 2-24 Chebyshev (a) and Sylvester-Kempe (b) Mechanisms Adapted from Kempe, *How to Draw a Straight Line*, Macmillan: London, 1877

- 2-23 Find the rotatability of each loop of the mechanisms in Figure P2-4e, f, and g. Scale the diagrams for dimensions.
- *2-24 Find the *DOF* of the mechanisms in Figure P2-5. Scale the diagrams for dimensions.
- 2-25 Find the *DOF* of the ice tongs in Figure P2-6.
- When operating them to grab the ice block.
 - When clamped to the ice block but before it is picked up (ice grounded).
 - When the person is carrying the ice block with the tongs.
- *2-26 Find the *DOF* of the automotive throttle mechanism in Figure P2-7.
- 2-27 Sketch a kinematic diagram of the scissors jack shown in Figure P2-8 and determine its *DOF*. Describe how it works.
- 2-28 Find the *DOF* of the corkscrew in Figure P2-9.

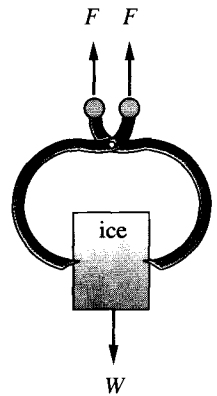


FIGURE P2-6

Problem 2-25

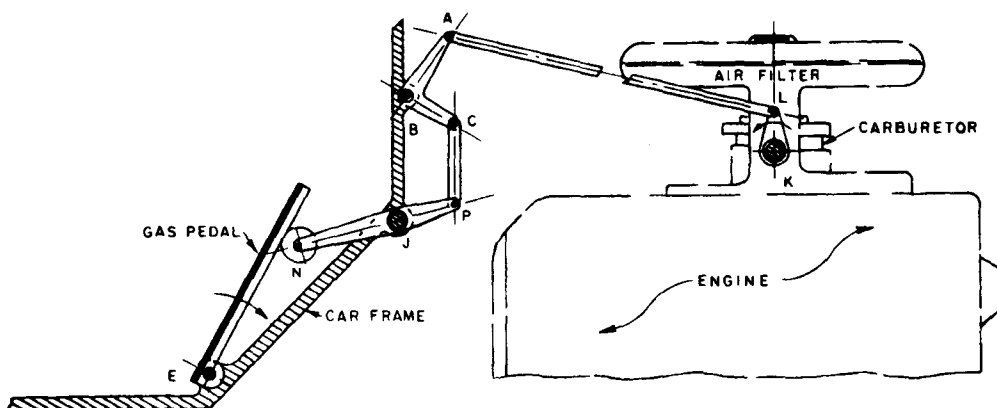


FIGURE P2-7

Problem 2-26. Adapted from P. H. Hill and W. P. Rule. (1960) *Mechanisms: Analysis and Design*, with permission.

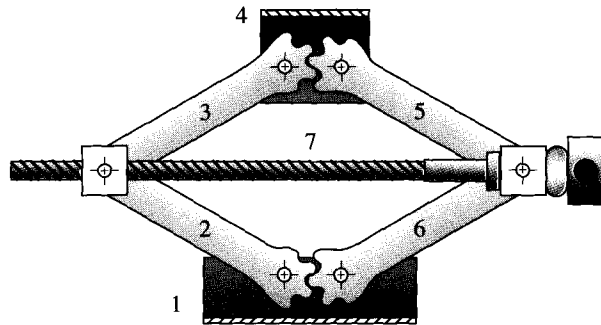


FIGURE P2-8

Problem 2-27

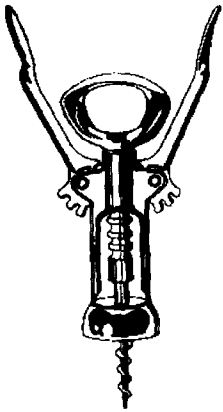


FIGURE 2-9

Problem 2-28

- 2-29 Figure P2-10 shows Watt's sun and planet drive that he used in his steam engine. The beam 2 is driven in oscillation by the piston of the engine. The planet gear is fixed rigidly to link 3 and its center is guided in the fixed track 1. The output rotation is taken from the sun gear 4. Sketch a kinematic diagram of this mechanism and determine its *DOF*. Can it be classified by the Barker scheme? If so, what Barker class and subclass is it?
- 2-30 Figure P2-11 shows a bicycle handbrake lever assembly. Sketch a kinematic diagram of this device and draw its equivalent linkage. Determine its *DOF*. Hint: Consider the flexible cable to be a link.

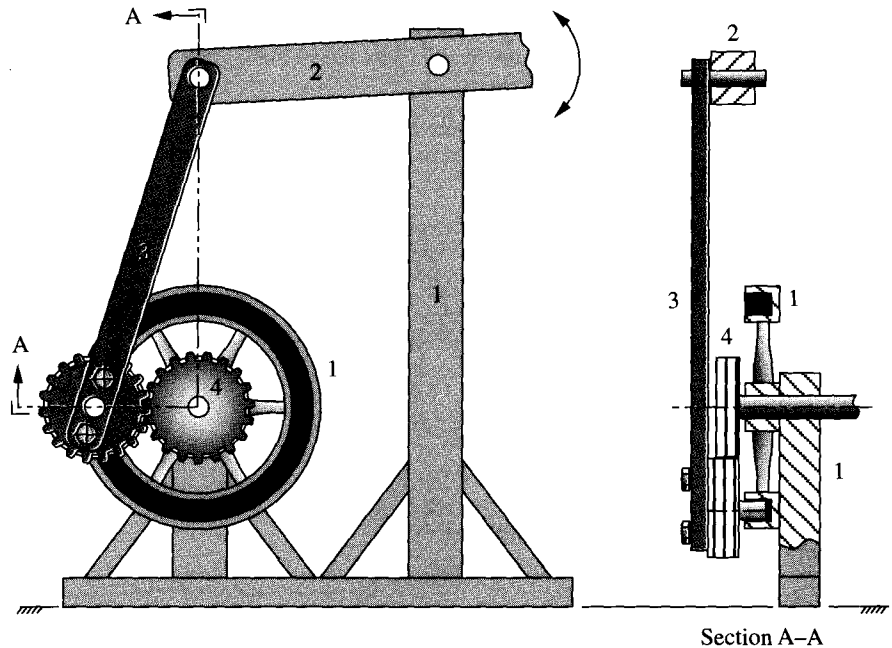
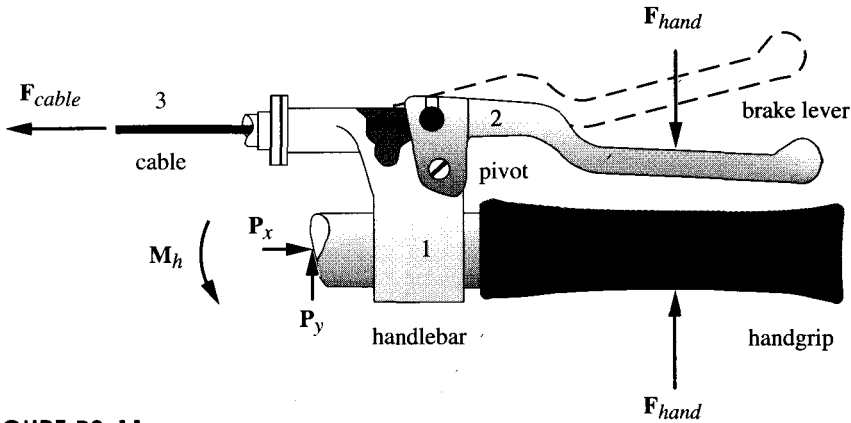


FIGURE P2-10

Problem 2-29 James Watt's sun and planet drive

**FIGURE P2-11**

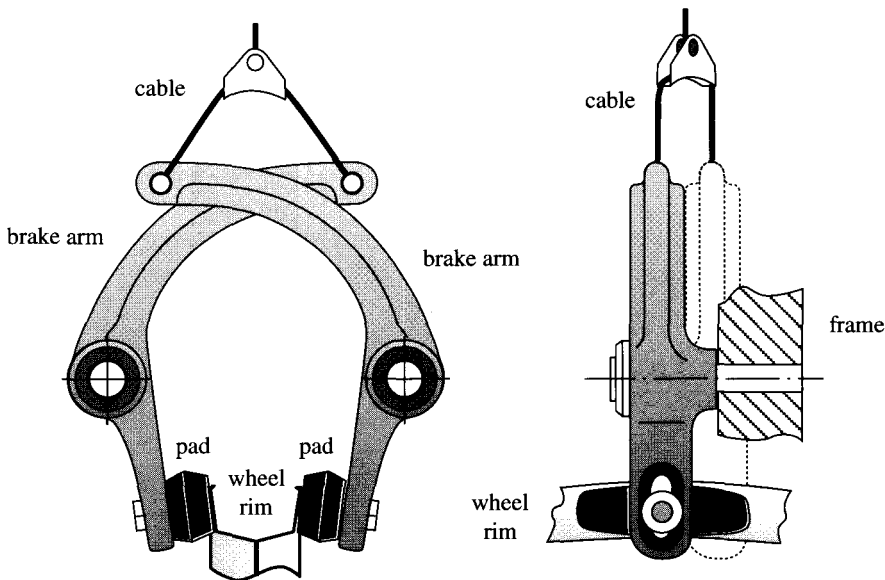
Problem 2-30 Bicycle hand brake lever assembly

2-31 Figure P2-12 shows a bicycle brake caliper assembly. Sketch a kinematic diagram of this device and draw its equivalent linkage. Determine its *DOF* under two conditions.

- Brake pads not contacting the wheel rim.
- Brake pads contacting the wheel rim.

Hint: Consider the flexible cables to be replaced by forces in this case.

2-32 Find the *DOF*, the Grashof condition, and the Barker classification of the mechanism in Figure P2-13.

**FIGURE P2-12**

Problem 2-31 Bicycle brake caliper assembly

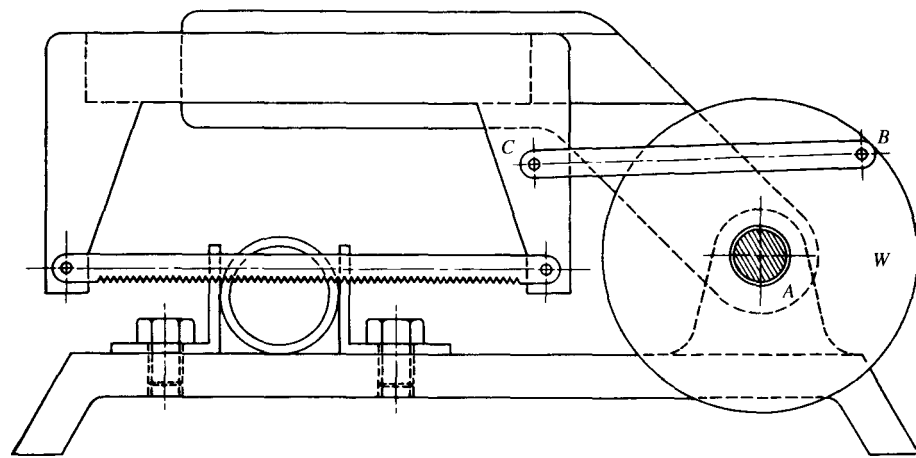


FIGURE P2-15
Problem 2-34 Power hacksaw. Adapted from P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

2-36 Sketch the equivalent linkage for the cam and follower mechanism in Figure P2-17 in the position shown. Show that it has the same *DOF* as the original mechanism.

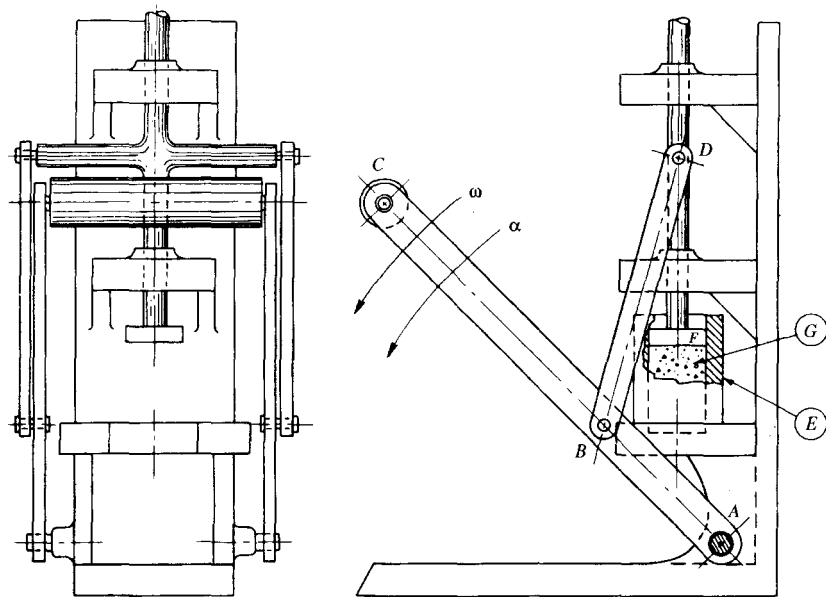


FIGURE P2-16
Problem 2-35 Powder compacting press. Adapted from P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

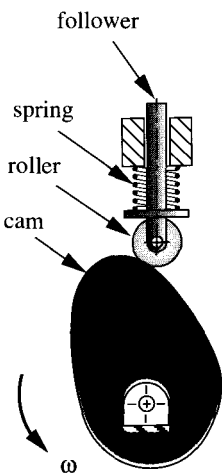
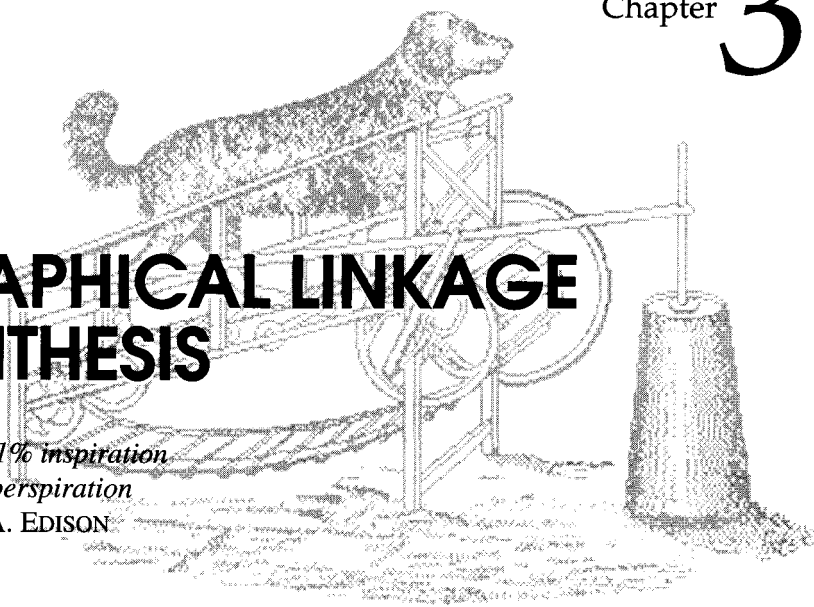


FIGURE P2-17
Problem 2-36



GRAPHICAL LINKAGE SYNTHESIS

*Genius is 1% inspiration
and 99% perspiration*

THOMAS A. EDISON

3.0 INTRODUCTION

Most engineering design practice involves a combination of synthesis and analysis. Most engineering courses deal primarily with analysis techniques for various situations. However, one cannot analyze anything until it has been synthesized into existence. Many machine design problems require the creation of a device with particular motion characteristics. Perhaps you need to move a tool from position *A* to position *B* in a particular time interval. Perhaps you need to trace out a particular path in space to insert a part into an assembly. The possibilities are endless, but a common denominator is often the need for a linkage to generate the desired motions. So, we will now explore some simple synthesis techniques to enable you to create potential linkage design solutions for some typical kinematic applications.

3.1 SYNTHESIS

QUALITATIVE SYNTHESIS means *the creation of potential solutions in the absence of a well-defined algorithm which configures or predicts the solution*. Since most real design problems will have many more unknown variables than you will have equations to describe the system's behavior, you cannot simply solve the equations to get a solution. Nevertheless you must work in this fuzzy context to create a potential solution and to also judge its **quality**. You can then analyze the proposed solution to determine its viability, and iterate between synthesis and analysis, as outlined in the **design** process, until you are satisfied with the result. Several tools and techniques exist to assist you in this process. The traditional tool is the drafting board, on which you layout, to scale,

multiple orthographic views of the design, and investigate its motions by drawing arcs, showing multiple positions, and using transparent, movable overlays. *Computer-aided drafting (CAD)* systems can speed this process to some degree, but you will probably find that the quickest way to get a sense of the quality of your linkage design is to model it, to scale, in cardboard or drafting Mylar® and see the motions directly.

Other tools are available in the form of computer programs such as FOURBAR, FIVEBAR, SIXBAR, SLIDER, DYNACAM, ENGINE, and MATRIX (all included with this text), some of which do synthesis, but these are mainly analysis tools. They can analyze a trial mechanism solution so rapidly that their dynamic graphical output gives almost instantaneous visual feedback on the quality of the design. Commercially available programs such as *Working Model** also allow rapid analysis of a proposed mechanical design. The process then becomes one of **qualitative design by successive analysis** which is really *an iteration between synthesis and analysis*. Very many trial solutions can be examined in a short time using these *Computer-aided engineering (CAE)* tools. We will develop the mathematical solutions used in these programs in subsequent chapters in order to provide the proper foundation for understanding their operation. But, if you want to try these programs to reinforce some of the concepts in these early chapters, you may do so. Appendix A is a manual for the use of these programs, and it can be read at any time. Reference will be made to program features which are germane to topics in each chapter, as they are introduced. Data files for input to these computer programs are also provided on disk for example problems and figures in these chapters. The data file names are noted near the figure or example. The student is encouraged to input these sample files to the programs in order to observe more dynamic examples than the printed page can provide. These examples can be run by merely accepting the defaults provided for all inputs.

TYPE SYNTHESIS refers to *the definition of the proper type of mechanism best suited to the problem* and is a form of qualitative synthesis.^t This is perhaps the most difficult task for the student as it requires some experience and knowledge of the various types of mechanisms which exist and which also may be feasible from a performance and manufacturing standpoint. As an example, assume that the task is to design a device to track the straight-line motion of a part on a conveyor belt and spray it with a chemical coating as it passes by. This has to be done at high, constant speed, with good accuracy and repeatability, and it must be reliable. Moreover, the solution must be inexpensive. Unless you have had the opportunity to see a wide variety of mechanical equipment, you might not be aware that this task could conceivably be accomplished by any of the following devices:

- A straight-line linkage
- A cam and follower
- An air cylinder
- A hydraulic cylinder
- A robot
- A solenoid

Each of these solutions, while possible, may not be optimal or even practical. More detail needs to be known about the problem to make that judgment, and that detail will come from the research phase of the design process. The straight-line linkage may prove to be too large and to have undesirable accelerations; the cam and follower will be expensive, though accurate and repeatable. The air cylinder itself is inexpensive but is noisy and unreliable. The hydraulic cylinder is more expensive, as is the robot. The so-

* The student version of *Working Model* is included on CD-ROM with this book. The professional version is available from Knowledge Revolution Inc., San Mateo CA 94402, (800) 766-6615.

^t A good discussion of type synthesis and an extensive bibliography on the topic can be found in Olson, D.G., et al. (1985). "A Systematic Procedure for Type Synthesis of Mechanisms with Literature Review," *Mechanism and Machine Theory*, 20(4), pp. 285-295.

lenoid, while cheap, has high impact loads and high impact velocity. So, you can see that the choice of device type can have a large effect on the quality of the design. A poor choice at the type synthesis stage can create insoluble problems later on. The design might have to be scrapped after completion, at great expense. Design is essentially an exercise in trade-offs. Each proposed type of solution in this example has good and bad points. Seldom will there be a clear-cut, obvious solution to a real engineering design problem. It will be your job as a design engineer to balance these conflicting features and find a solution which gives the best trade-off of functionality against cost, reliability, and all other factors of interest. Remember, *an engineer can do, with one dollar, what any fool can do for ten dollars*. Cost is always an important constraint in engineering design.

QUANTITATIVE SYNTHESIS, OR ANALYTICAL SYNTHESIS means the generation of one or more solutions of a particular type which you know to be suitable to the problem, and more importantly, one for which there is a synthesis algorithm defined. As the name suggests, this type of solution can be quantified, as some set of equations exists which will give a numerical answer. Whether that answer is a good or suitable one is still a matter for the judgment of the designer and requires analysis and iteration to optimize the design. Often the available equations are fewer than the number of potential variables, in which case you must assume some reasonable values for enough unknowns to reduce the remaining set to the number of available equations. Thus some qualitative judgment enters into the synthesis in this case as well. Except for very simple cases, a CAE tool is needed to do quantitative synthesis. One example of such a tool is the program LINCAGES,* by A. Erdman et al., of the University of Minnesota [1] which solves the three-position and four-position linkage synthesis problems. The computer programs provided with this text also allow you to do three-position analytical synthesis and general linkage design by successive analysis. The fast computation of these programs allows one to analyze the performance of many trial mechanism designs in a short time and promotes rapid iteration to a better solution.

DIMENSIONAL SYNTHESIS of a linkage is the determination of the proportions (lengths) of the links necessary to accomplish the desired motions and can be a form of quantitative synthesis if an algorithm is defined for the particular problem, but can also be a form of qualitative synthesis if there are more variables than equations. The latter situation is more common for linkages. (Dimensional synthesis of cams is quantitative.) Dimensional synthesis assumes that, through type synthesis, you have already determined that a linkage (or a cam) is the most appropriate solution to the problem. This chapter discusses graphical dimensional synthesis of linkages in detail. Chapter 5 presents methods of analytical linkage synthesis, and Chapter 8 presents cam synthesis.

3.2 FUNCTION, PATH, AND MOTION GENERATION

FUNCTION GENERATION is defined as the correlation of an input motion with an output motion in a mechanism. A function generator is conceptually a "black box" which delivers some predictable output in response to a known input. Historically, before the advent of electronic computers, mechanical function generators found wide application in artillery rangefinders and shipboard gun aiming systems, and many other tasks. They are, in fact, mechanical analog computers. The development of inexpensive digital electronic microcomputers for control systems coupled with the availability of compact

* Available from
Knowledge Revolution Inc.,
San Mateo CA 94402,
(800) 766-6615.

servo and stepper motors has reduced the demand for these mechanical function generator linkage devices. Many such applications can now be served more economically and efficiently with electromechanical devices.* Moreover, the computer-controlled electromechanical function generator is programmable, allowing rapid modification of the function generated as demands change. For this reason, while presenting some simple examples in this chapter and a general, analytical design method in Chapter 5, we will not emphasize mechanical linkage function generators in this text. Note however that the cam-follower system, discussed extensively in Chapter 8, is in fact a form of mechanical function generator, and it is typically capable of higher force and power levels per dollar than electromechanical systems.

PATH GENERATION is defined as *the control of a point in the plane such that it follows some prescribed path*. This is typically accomplished with at least four bars, wherein a point on the coupler traces the desired path. Specific examples are presented in the section on coupler curves below. Note that no attempt is made in path generation to control the orientation of the link which contains the point of interest. However, it is common for the timing of the arrival of the point at particular locations along the path to be defined. This case is called *path generation with prescribed timing* and is analogous to function generation in that a particular output function is specified. Analytical path and function generation will be dealt with in Chapter 5.

MOTION GENERATION is defined as *the control of a line in the plane such that it assumes some prescribed set of sequential positions*. Here orientation of the link containing the line is important. This is a more general problem than path generation, and in fact, path generation is a subset of motion generation. An example of a motion generation problem is the control of the "bucket" on a bulldozer. The bucket must assume a set of positions to dig, pick up, and dump the excavated earth. Conceptually, the motion of a line, painted on the side of the bucket, must be made to assume the desired positions. A linkage is the usual solution.

PLANAR MECHANISMS VERSUS SPATIAL MECHANISMS The above discussion of controlled movement has assumed that the motions desired are planar (2-D). We live in a three-dimensional world, however, and our mechanisms must function in that world. Spatial mechanisms are 3-D devices. Their design and analysis is much more complex than that of planar mechanisms, which are 2-D devices. The study of spatial mechanisms is beyond the scope of this introductory text. Some references for further study are in the bibliography to this chapter. However, the study of planar mechanisms is not as practically limiting as it might first appear since many devices in three dimensions are constructed of multiple sets of 2-D devices coupled together. An example is any folding chair. It will have some sort of linkage in the left side plane which allows folding. There will be an identical linkage on the right side of the chair. These two XY planar linkages will be connected by some structure along the Z direction, which ties the two planar linkages into a 3-D assembly. Many real mechanisms are arranged in this way, as duplicate planar linkages, displaced in the Z direction in parallel planes and rigidly connected. When you open the hood of a car, take note of the hood hinge mechanism. It will be duplicated on each side of the car. The hood and the car body tie the two planar linkages together into a 3-D assembly. Look and you will see many other such examples of assemblies of planar linkages into 3-D configurations. So, the 2-D techniques of synthesis and analysis presented here will prove to be of practical value in designing in 3-D as well.

* It is worth noting that the day is long past when a mechanical engineer can be content to remain ignorant of electronics and electromechanics. Virtually all modern machines are controlled by electronic devices. Mechanical engineers must understand their operation.

3.3 LIMITING CONDITIONS

The manual, graphical, dimensional synthesis techniques presented in this chapter and the computerizable, analytical synthesis techniques presented in Chapter 5 are reasonably rapid means to obtain a trial solution to a motion control problem. Once a potential solution is found, it must be evaluated for its quality. There are many criteria which may be applied. In later chapters, we will explore the analysis of these mechanisms in detail. However, one does not want to expend a great deal of time analyzing, in great detail, a design which can be shown to be inadequate by some simple and quick evaluations.

TOGGLE One important test can be applied within the synthesis procedures described below. You need to check that the linkage can in fact reach all of the specified design positions without encountering a limit or toggle position, also called a stationary configuration. Linkage synthesis procedures often only provide that the particular positions specified will be obtained. They say nothing about the linkage's behavior between those positions. Figure 3-1a shows a non-Grashof fourbar linkage in an arbitrary position CD (dashed lines), and also in its two toggle positions, C_1D_1 (solid black lines) and C_2D_2 (solid red lines). The toggle positions are determined by the colinearity of two of the moving links. A fourbar double- or triple-rocker mechanism will have at least two of these toggle positions in which the linkage assumes a triangular configuration. When in a triangular (toggle) position, it will not allow further input motion in one direction from one of its rocker links (either of link 2 from position C_1D_1 or link 4 from position C_2D_2). The other rocker will then have to be driven to get the linkage out of toggle. A Grashof fourbar crank-rocker linkage will also assume two toggle positions as shown in Figure 3-1b, when the shortest link (crank O_2C) is colinear with the coupler CD (link 3), either extended colinear ($O_2C_2D_2$) or overlapping colinear ($O_2C_1D_1$). It cannot be back driven from the rocker O_4D (link 4) through these colinear positions, but when the crank O_2C (link 2) is driven, it will carry through both toggles because it is Grashof. Note that these toggle positions also define the limits of motion of the driven rocker (link 4), at which its angular velocity will go through zero. Use program FOURBAR to read the data files F03-01AABR and F03-1bABR and animate these examples.

After synthesizing a double- or triple-rocker solution to a multiposition (motion generation) problem, you must check for the presence of toggle positions between your

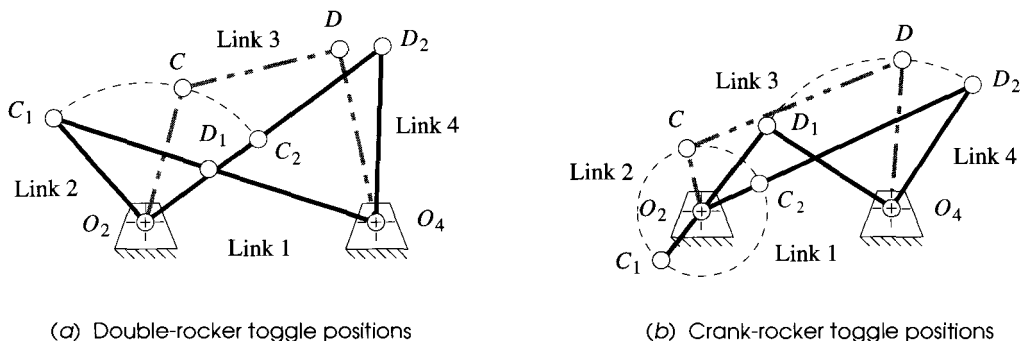


FIGURE 3-1

Linkages in toggle

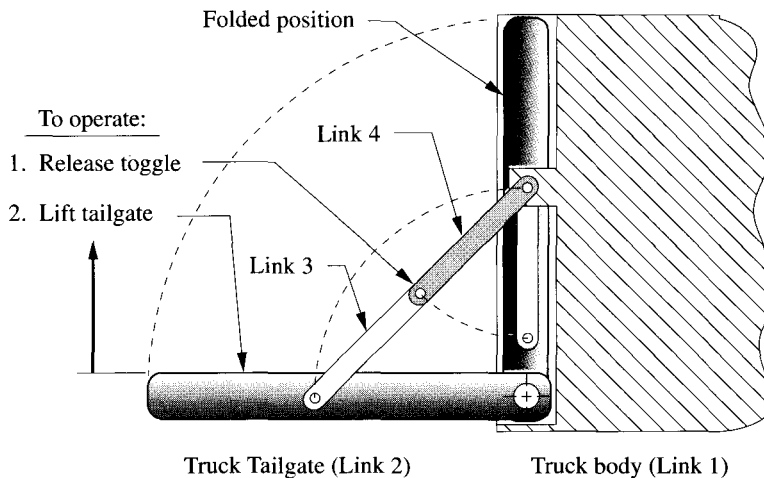


FIGURE 3-2

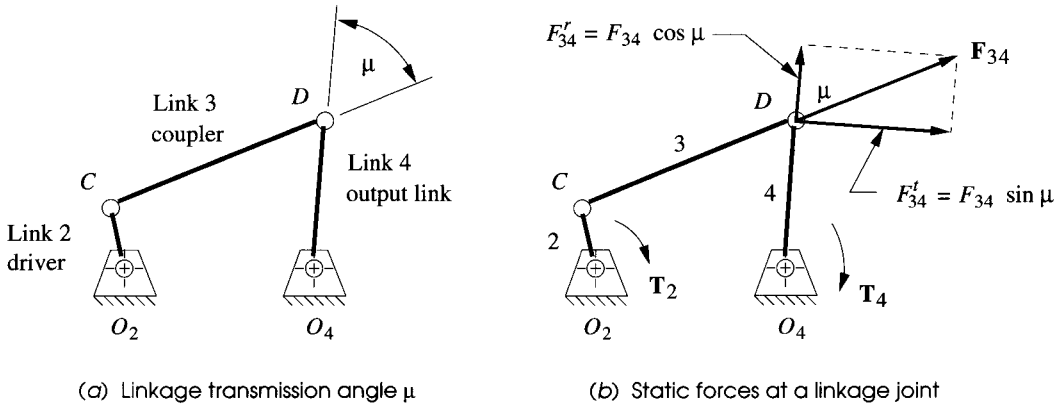
Deltoid toggle linkage used to control truck tailgate motion

design positions. An easy way to do this is with a cardboard model of the linkage design. A CAE tool such as FOURBAR or Working Model will also check for this problem. It is important to realize that a toggle condition is only undesirable if it is preventing your linkage from getting from one desired position to the other. In other circumstances the toggle is very useful. It can provide a self-locking feature when a linkage is moved slightly beyond the toggle position and against a fixed stop. Any attempt to reverse the motion of the linkage then causes it merely to jam harder against the stop. It must be manually pulled "over center," out of toggle, before the linkage will move. You have encountered many examples of this application, as in card table or ironing board leg linkages and also pickup truck or station wagon tailgate linkages. An example of such a toggle linkage is shown in Figure 3-2. It happens to be a special-case Grashof linkage in the deltoid configuration (see also Figure 2-17d, p. 49), which provides a locking toggle position when open, and folds on top of itself when closed, to save space. We will analyze the toggle condition in more detail in a later chapter.

TRANSMISSION ANGLE Another useful test that can be very quickly applied to a linkage design to judge its quality is the measurement of its transmission angle. This can be done analytically, graphically on the drawing board, or with the cardboard model for a rough approximation. (Extend the links beyond the pivot to measure the angle.) The **transmission angle** is shown in Figure 3-3a and is defined as *the angle between the output link and the coupler*.^{*} It is usually taken as the *absolute value of the acute angle of the pair of angles at the intersection of the two links and varies continuously from some minimum to some maximum value as the linkage goes through its range of motion*. It is a measure of the quality of force and velocity transmission at the joint.[†] Note in Figure 3-2 that the linkage cannot be moved from the open position shown by any force applied to the tailgate, link 2, since the transmission angle is then between links 3 and 4 and is zero at that position. But a force applied to link 4 as the input link will move it. The transmission angle is now between links 3 and 2 and is 45 degrees.

^{*} As defined by Alt. [2]

[†] The transmission angle has limited application. It only predicts the quality of force or torque transmission if the input and output links are pivoted to ground. If the output force is taken from a floating link (coupler), then the transmission angle is of no value. A different index of merit called the joint force index (JFI) is presented in Chapter 11 which discusses force analysis in linkages. (See Section 11.12 on p. 554.) The JFI is useful for situations in which the output link is floating as well as for giving the same kind of information when the output is taken from a link rotating against the ground. However, the JFI requires that a complete force analysis of the linkage be done, whereas the transmission angle is determined from linkage geometry alone.

**FIGURE 3-3**

Transmission angle in the fourbar linkage

* Alt, [2] who defined the transmission angle, recommended keeping $\mu_{\min} > 40^\circ$. But it can be argued that at higher speeds, the momentum of the moving elements and/or the addition of a flywheel will carry a mechanism through locations of poor transmission angle. The most common example is the back-driven slider crank (as used in internal combustion engines) which has $\mu = 0$ twice per revolution. Also, the transmission angle is only critical in a fourbar linkage when the rocker is the output link on which the working load impinges. If the working load is taken by the coupler rather than by the rocker, then minimum transmission angles less than 40° may be viable. A more definitive way to qualify a mechanism's dynamic function is to compute the variation in its required driving torque. Driving torque and flywheels are addressed in Chapter II. A joint force index (JFI) can also be calculated. (See footnote on p. 81.)

Figure 3-3b shows a torque T_2 applied to link 2. Even before any motion occurs, this causes a static, colinear force F_{34} to be applied by link 3 to link 4 at point D . Its radial and tangential components F_{34}^r and F_{34}^t are resolved parallel and perpendicular to link 4, respectively. Ideally, we would like all of the force F_{34} to go into producing output torque T_4 on link 4. However, only the tangential component creates torque on link 4. The radial component F_{34}^r provides only tension or compression in link 4. This radial component only increases pivot friction and does not contribute to the output torque. Therefore, the optimum value for the transmission angle is 90° . When μ is less than 45° the radial component will be larger than the tangential component. Most machine designers try to keep the minimum transmission angle above about 40° to promote smooth running and good force transmission. However, if in your particular design there will be little or no external force or torque applied to link 4, you may be able to get away with even lower values of μ .^{*} The transmission angle provides one means to judge the quality of a newly synthesized linkage. If it is unsatisfactory, you can iterate through the synthesis procedure to improve the design. We will investigate the transmission angle in more detail in later chapters.

3.4 DIMENSIONAL SYNTHESIS

Dimensional synthesis of a linkage is the determination of the proportions (lengths) of the links necessary to accomplish the desired motions. This section assumes that, through type synthesis, you have determined that a linkage is the most appropriate solution to the problem. Many techniques exist to accomplish this task of dimensional synthesis of a fourbar linkage. The simplest and quickest methods are graphical. These work well for up to three design positions. Beyond that number, a numerical, analytical synthesis approach as described in Chapter 5, using a computer, is usually necessary.

Note that the principles used in these graphical synthesis techniques are simply those of euclidean geometry. The rules for bisection of lines and angles, properties of parallel

and perpendicular lines, and definitions of arcs, etc., are all that are needed to generate these linkages. Compass, protractor, and rule are the only tools needed for graphical linkage synthesis. Refer to any introductory (high school) text on geometry if your geometric theorems are rusty.

Two-Position Synthesis

Two-position synthesis subdivides into two categories: rocker output (pure rotation) and coupler output (complex motion). Rocker output is most suitable for situations in which a Grashof crank-rocker is desired and is, in fact, a trivial case of *function generation* in which the output function is defined as two discrete angular positions of the rocker. Coupler output is more general and is a simple case of *motion generation* in which two positions of a line in the plane are defined as the output. This solution will frequently lead to a triple-rocker. However, the fourbar triple-rocker can be motor driven by the addition of a dyad (two-bar chain), which makes the final result a Watt's sixbar containing a Grashof fourbar subchain. We will now explore the synthesis of each of these types of solution for the two-position problem.

EXAMPLE 3-1

Rocker Output - Two Positions with Angular Displacement. (Function Generation)

Problem: Design a fourbar Grashof crank-rocker to give 45° of rocker rotation with equal time forward and back, from a constant speed motor input.

Solution: (see Figure 3-4)

- 1 Draw the output link O_1V in both extreme positions, B_1 and B_2 in any convenient location, such that the desired angle of motion θ_4 is subtended.
- 2 Draw the chord B_1B_2 and extend it in any convenient direction.
- 3 Select a convenient point O_2 on line B_1B_2 extended.
- 4 Bisect line segment B_1B_2 , and draw a circle of that radius about O_2 .
- 5 Label the two intersections of the circle and B_1B_2 extended, A_1 and A_2 .
- 6 Measure the length of the coupler as A_1 to B_1 or A_2 to B_2 .
- 7 Measure ground length 1, crank length 2, and rocker length 4.
- 8 Find the Grashof condition. If non-Grashof, redo steps 3 to 8 with O_2 further from O_4 .
- 9 Make a cardboard model of the linkage and articulate it to check its function and its transmission angles.
- 10 You can input the file F03-04.4br to program FOURBAR to see this example come alive.

Note several things about this synthesis process. We started with the output end of the system, as it was the only aspect defined in the problem statement. We had to make

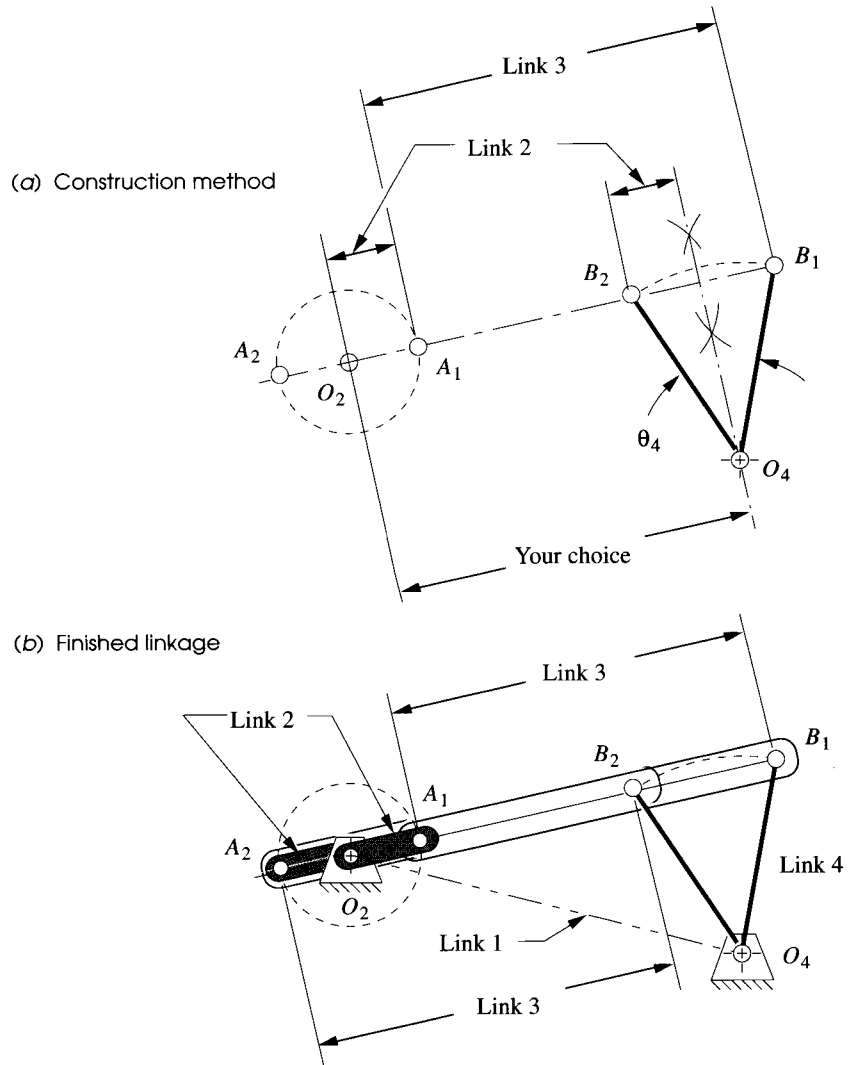


FIGURE 3-4

Two-position function synthesis with rocker output (non-quick-return)

many quite arbitrary decisions and assumptions to proceed because there were many more variables than we could have provided "equations" for. We are frequently forced to make "free choices" of "a convenient angle or length." These free choices are actually definitions of design parameters. A poor choice will lead to a poor design. Thus these are **qualitative synthesis** approaches and require an iterative process, even for this simple an example. The first solution you reach will probably not be satisfactory, and several attempts (iterations) should be expected to be necessary. As you gain more experience in designing kinematic solutions you will be able to make better choices for these

design parameters with fewer iterations. The value of making a simple model of your design cannot be overstressed! You will get the *most insight* into your design's quality for the *least effort* by making, articulating, and studying the model. These general observations will hold for most of the linkage synthesis examples presented.

EXAMPLE 3-2

Rocker Output - Two Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move link CD from position C_1D_1 to C_2D_2 .

Solution: (see Figure 3-5)

- 1 Draw the link CD in its two desired positions, C_1D_1 and C_2D_2 , in the plane as shown.
- 2 Draw construction lines from point C_1 to C_2 and from point D_1 to D_2 .
- 3 Bisect line C_1C_2 and line D_1D_2 and extend their perpendicular bisectors to intersect at O_4 . Their intersection is the **rotopole**.
- 4 Select a convenient radius and draw an arc about the rotopole to intersect both lines O_4C_1 and O_4C_2 . Label the intersections B_1 and B_2 .
- 5 Do steps 2 to 8 of Example 3-1 above to complete the linkage.
- 6 Make a cardboard model of the linkage and articulate it to check its function and its transmission angles.

Note that Example 3-2 reduces to the method of Example 3-1 once the **rotopole** is found. Thus a link represented by a line in complex motion can be reduced to the simpler problem of pure rotation and moved to any two positions in the plane as the rocker on a fourbar linkage. The next example moves the same link through the same two positions as the coupler of a fourbar linkage.

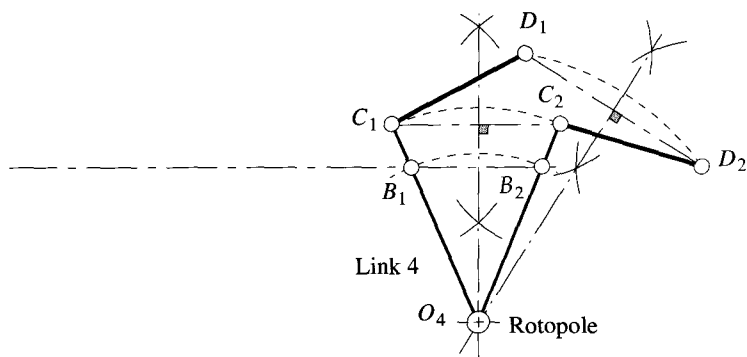
EXAMPLE 3-3

Coupler Output - Two Positions with Complex Displacement. (Motion Generation)

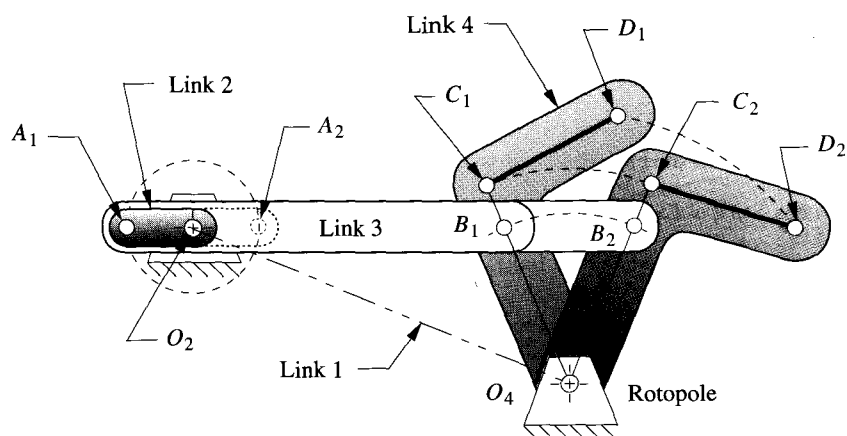
Problem: Design a fourbar linkage to move the link CD shown from position $C)D)$ to C_2D_2 (with moving pivots at C and D).

Solution: (see Figure 3-6)

- 1 Draw the link CD in its two desired positions, $C)D)$ and C_2D_2 , in the plane as shown.
- 2 Draw construction lines from point $C)$ to C_2 and from point $D)$ to D_2 .
- 3 Bisect line $C)C_2$ and line $D)D_2$ and extend the perpendicular bisectors in convenient directions. The rotopole will not be used in this solution.



(a) Finding the rotapole for Example 3-2

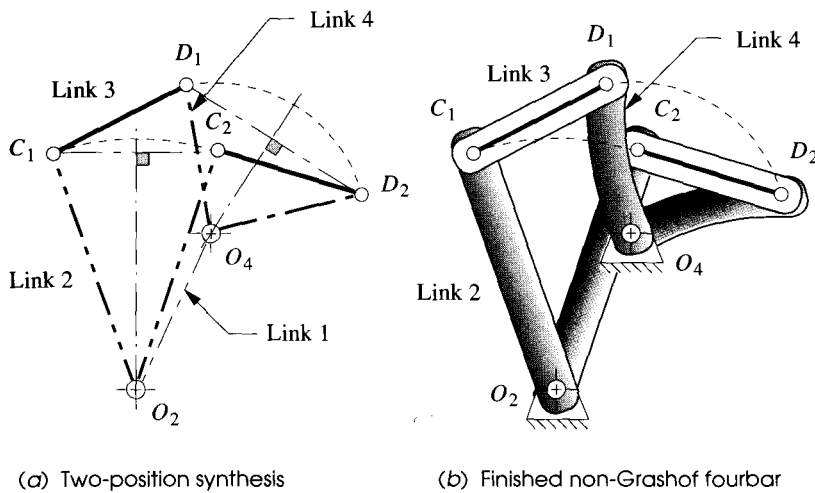


(b) Constructing the linkage by the method in Example 3-1

FIGURE 3-5

Two-position motion synthesis with rocker output (non-quick-return)

- 4 Select any convenient point on each bisector as the fixed pivots O_2 and O_4 , respectively.
- 5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.
- 6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.
- 7 Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.
- 8 Construct a cardboard model and check its function to be sure it can get from the initial to final position without encountering any limit (toggle) positions.
- 9 Check transmission angles.



(a) Two-position synthesis

(b) Finished non-Grashof fourbar

FIGURE 3-6

Two-position motion synthesis with coupler output

Input file F03-06.4br to program FOURBAR to see Example 3-3. Note that this example had nearly the same problem statement as Example 3-2, but the solution is quite different. Thus a link can also be moved to any two positions in the plane as the coupler of a fourbar linkage, rather than as the rocker. However, to limit its motions to those two coupler positions as extrema, two additional links are necessary. These additional links can be designed by the method shown in Example 3-4 and Figure 3-7.

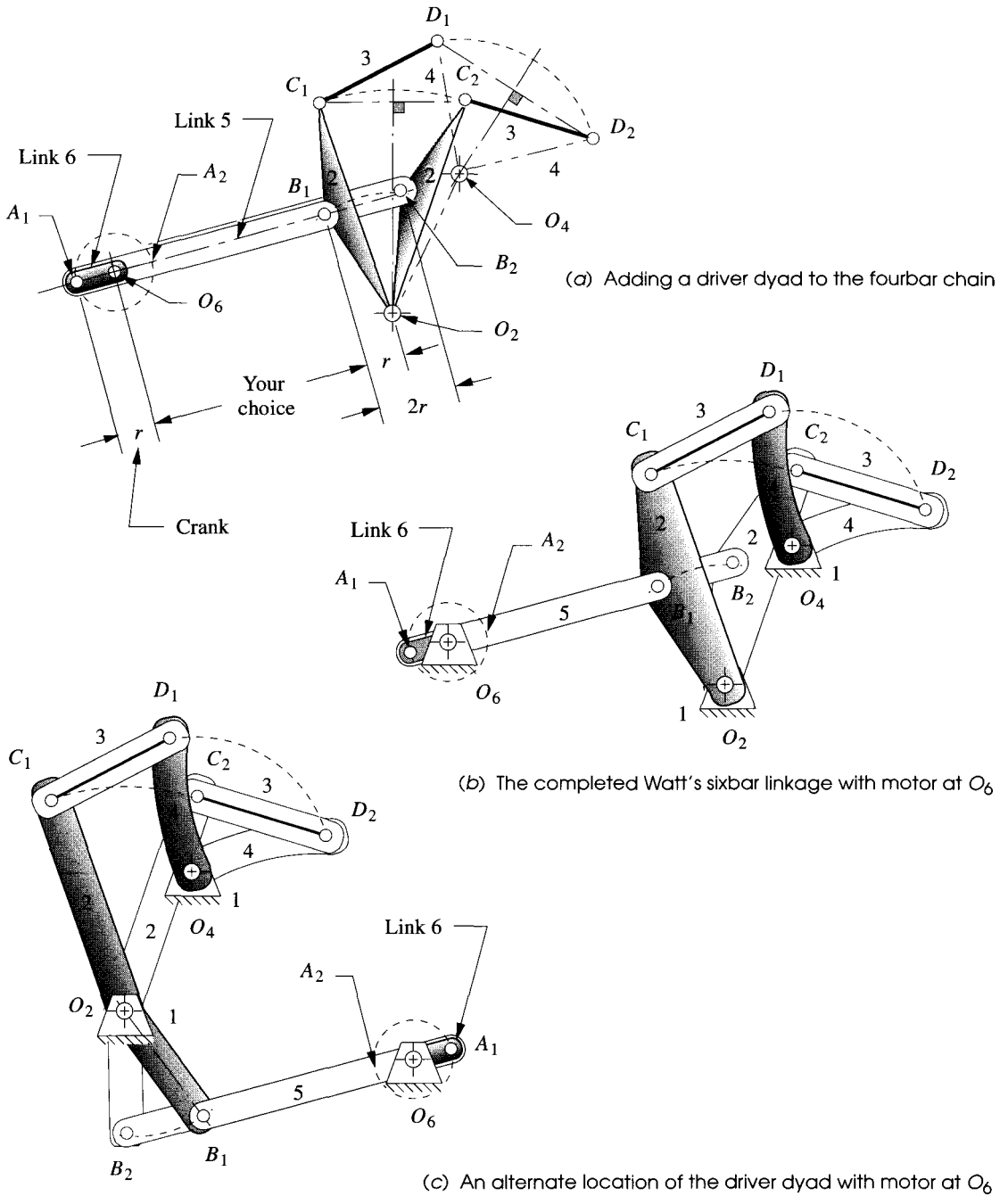
EXAMPLE 3-4

Adding a Dyad (Twobar Chain) to Control Motion in Example 3-3.

Problem: Design a **dyad** to control and limit the extremes of motion of the linkage in Example 3-3 to its two design positions.

Solution: (see Figure 3-7a)

- 1 Select a convenient point on link 2 of the linkage designed in Example 3-3. Note that it need not be on the line O_2C_1 . Label this point B_1 .
- 2 Draw an arc about center O_2 through B_1 to intersect the corresponding line O_2B_2 in the second position of link 2. Label this point B_2 . The chord B_1B_2 provides us with the same problem as in Example 3-1.
- 3 Do steps 2 to 9 of Example 3-1 to complete the linkage, except add links 5 and 6 and center O_6 rather than links 2 and 3 and center O_2 . Link 6 will be the driver crank. The fourbar sub-chain of links O_6, A_1, B_1, O_2 must be a Grashof crank-rocker.

**FIGURE 3-7**

Driving a non-Grashof linkage with a dyad (non-quick-return)

Note that we have used the approach of Example 3-1 to add a dyad to serve as a *driver stage* for our existing fourbar. This results in a sixbar Watt's mechanism whose first stage is Grashof as shown in Figure 3-7b. Thus we can drive this with a motor on link 6. Note also that we can locate the motor center O_6 anywhere in the plane by judicious choice of point B_1 on link 2. If we had put B_1 below center O_2 , the motor would be to the right of links 2, 3, and 4 as shown in Figure 3-7c. There is *an infinity of driver dyads* possible which will drive any double-rocker assemblage of links. Input the files RB-07b.6br and F03-07c.6br to program SIXBAR to see Example 3-4 in motion for these two solutions.

Three-Position Synthesis with Specified Moving Pivots

Three-position synthesis allows the definition of three positions of a line in the plane and will create a fourbar linkage configuration to move it to each of those positions. This is a motion generation problem. The synthesis technique is a logical extension of the method used in Example 3-3 for two-position synthesis with coupler output. The resulting linkage may be of any Grashof condition and will usually require the addition of a dyad to control and limit its motion to the positions of interest. Compass, protractor, and rule are the only tools needed in this graphical method.

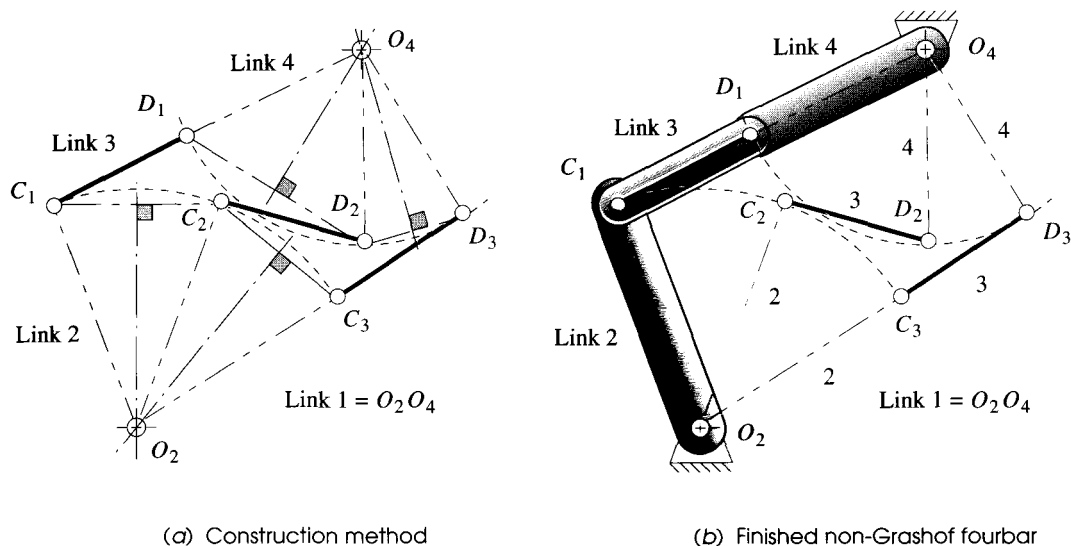
EXAMPLE 3-5

Coupler Output - 3 Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3 . Moving pivots are at C and D . Find the fixed pivot locations.

Solution: (see Figure 3-8)

- 1 Draw link CD in its three design positions C_1D_1 , C_2D_2 , C_3D_3 in the plane as shown.
- 2 Draw construction lines from point C_1 to C_2 and from point C_2 to C_3 .
- 3 Bisect line C_1C_2 and line C_2C_3 and extend their perpendicular bisectors until they intersect. Label their intersection O_2 .
- 4 Repeat steps 2 and 3 for lines D_1D_2 and D_2D_3 . Label the intersection O_4 .
- 5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.
- 6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.
- 7 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.
- 8 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.
- 9 Construct a driver dyad according to the method in Example 3-4 using an extension of link 3 to attach the dyad.

**FIGURE 3-8**

Three-position motion synthesis

Note that while a solution is usually obtainable for this case, it is possible that you may not be able to move the linkage continuously from one position to the next without disassembling the links and reassembling them to get them past a limiting position. That will obviously be unsatisfactory. In the particular solution presented in Figure 3-8, note that links 3 and 4 are in toggle at position one, and links 2 and 3 are in toggle at position three. In this case we will have to drive link 3 with a driver dyad, since any attempt to drive either link 2 or link 4 will fail at the toggle positions. No amount of torque applied to link 2 at position C_1 will move link 4 away from point D_1 and driving link 4 will not move link 2 away from position C_3 . Input the file F03-08.4br to program FOURBAR to see Example 3-5.

Three-Position Synthesis with Alternate Moving Pivots

Another potential problem is the possibility of an undesirable location of the fixed pivots O_2 and O_4 with respect to your packaging constraints. For example, if the fixed pivot for a windshield wiper linkage design ends up in the middle of the windshield, you may want to redesign it. Example 3-6 shows a way to obtain an alternate configuration for the same three-position motion as in Example 3-5. And, the method shown in Example 3-8 (ahead on p. 95) allows you to specify the location of the fixed pivots in advance and then find the locations of the moving pivots on link 3 that are compatible with those fixed pivots.

EXAMPLE 3-6

Coupler Output - Three Positions with Complex Displacement - Alternate Attachment Points for Moving Pivots. (Motion Generation)

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3 . Use different moving pivots than CD . Find the fixed pivot locations.

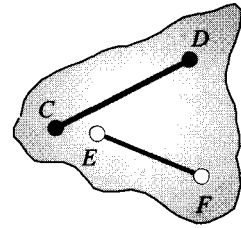
Solution: (see Figure 3-9)

- 1 Draw the link CD in its three desired positions C_1D_1 , C_2D_2 , C_3D_3 , in the plane as done in Example 3-5.
- 2 Define new attachment points E_1 and F_1 which have a fixed relationship between C_1D_1 and E_1F_1 within the link. Now use E_1F_1 to define the three positions of the link.
- 3 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3 .
- 4 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection O_2 .
- 5 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3 . Label the intersection O_4 .
- 6 Connect O_2 with E_1 and call it link 2. Connect O_4 with F_1 and call it link 4.
- 7 Line E_1F_1 is link 3. Line O_2O_4 is link 1.
- 8 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.
- 9 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions. If not, change locations of points E and F and repeat steps 3 to 9.
- 10 Construct a driver dyad acting on link 2 according to the method in Example 3-4.

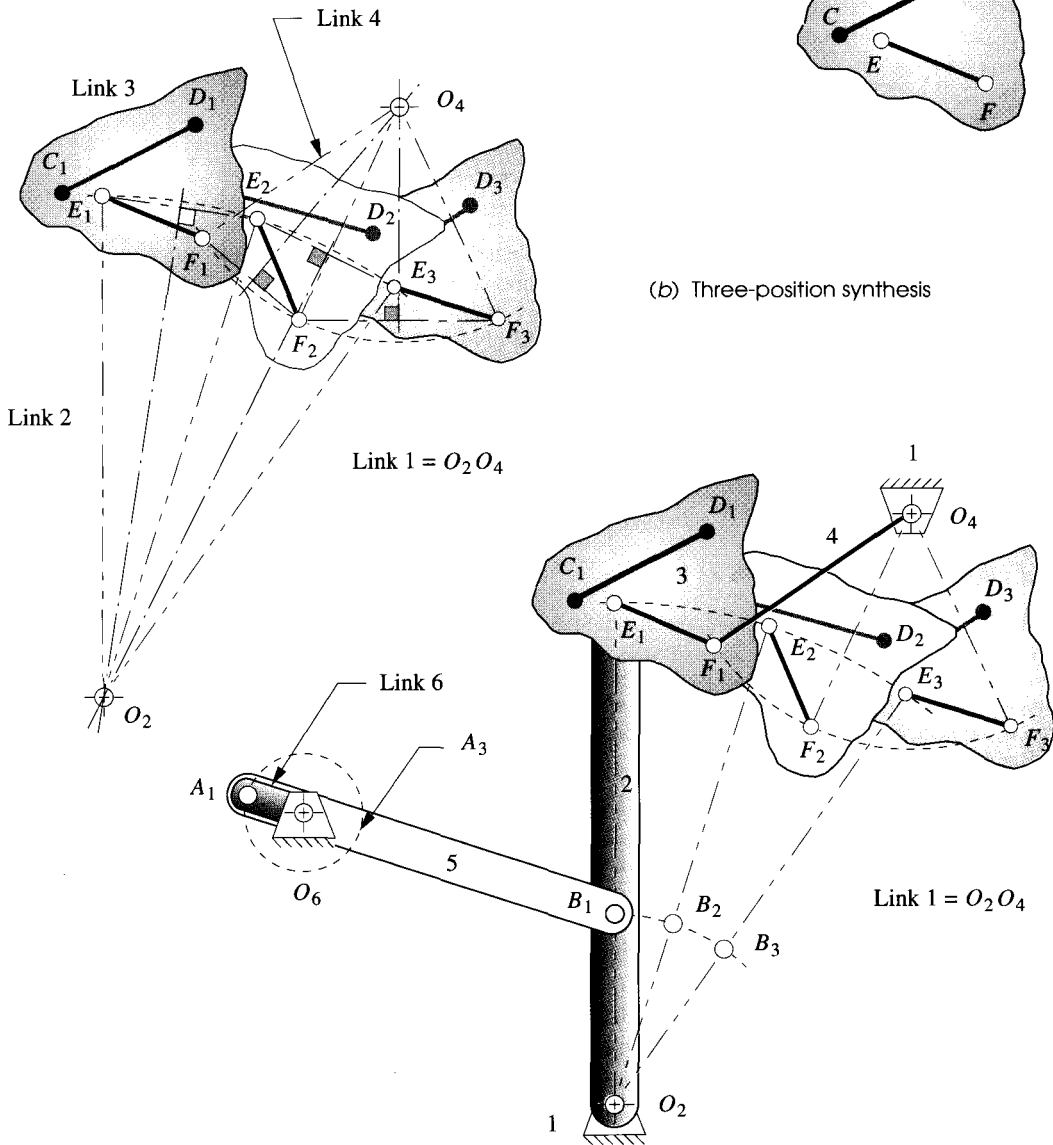
Note that the shift of the attachment points on link 3 from CD to EF has resulted in a shift of the locations of fixed pivots O_2 and O_4 as well. Thus they may now be in more favorable locations than they were in Example 3-5. It is important to understand that any two points on link 3, such as E and F , can serve to fully define that link as a rigid body, and that there is an infinity of such sets of points to choose from. While points C and D have some particular location in the plane which is defined by the linkage's function, points E and F can be anywhere on link 3, thus creating an infinity of solutions to this problem.

The solution in Figure 3-9 is different from that of Figure 3-8 in several respects. It avoids the toggle positions and thus can be driven by a dyad acting on one of the rockers, as shown in Figure 3-9c, and the transmission angles are better. However, the toggle positions of Figure 3-8 might actually be of value if a self-locking feature were desired. *Recognize that both of these solutions are to the same problem,* and the solution

(a) Alternate attachment points



(b) Three-position synthesis

(c) Completed Watt's sixbar linkage with motor at O_6 **FIGURE 3-9**

Three-position synthesis with alternate moving pivots

in Figure 3-8 is just a special case of that in Figure 3-9. Both solutions may be useful. Line CD moves through the same three positions with both designs. There is an infinity of other solutions to this problem waiting to be found as well. Input the file F03-09c.6br to program SrXBAR to see Example 3-6.

Three-Position Synthesis with Specified Fixed Pivots

Even though one can probably find an acceptable solution to the three-position problem by the methods described in the two preceding examples, it can be seen that the designer will have little direct control over the location of the fixed pivots since they are one of the results of the synthesis process. It is common for the designer to have some constraints on acceptable locations of the fixed pivots, since they will be limited to locations at which the ground plane of the package is accessible. It would be preferable if we could define the fixed pivot locations, as well as the three positions of the moving link, and then synthesize the appropriate attachment points, E and F , to the moving link to satisfy these more realistic constraints. The principle of **inversion** can be applied to this problem. Examples 3-5 and 3-6 showed how to find the required fixed pivots for three chosen positions of the moving pivots. Inverting this problem allows specification of the fixed pivot locations and determination of the required moving pivots for those locations. The first step is to find the three positions of the ground plane which correspond to the three desired coupler positions. This is done by **inverting the linkage** * as shown in Figure 3-10 and Example 3-7.

EXAMPLE 3-7

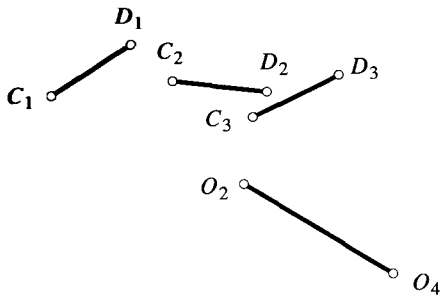
Three-Position Synthesis with Specified Fixed Pivots – Inverting the Three-Position Motion Synthesis Problem.

Problem: Invert a fourbar linkage which moves the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3 . Use specified fixed pivots O_2 and O_4 .

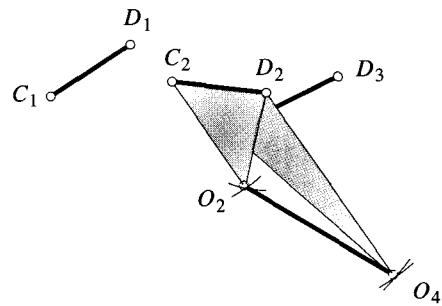
Solution: First find the inverted positions of the ground link corresponding to the three coupler positions specified. (See Figure 3-10.)

- 1 Draw the link CD in its three desired positions C_1D_1 , C_2D_2 , C_3D_3 , in the plane, as was done in Example 3-5 and as shown in Figure 3-10a.
- 2 Draw the ground link O_2O_4 in its desired position in the plane with respect to the first coupler position C_1D_1 as shown in Figure 3-10a.
- 3 Draw construction arcs from point C_2 to O_2 and from point D_2 to O_2 whose radii define the sides of triangle $C_2O_2D_2$. This defines the relationship of the fixed pivot O_2 to the coupler line CD in the second coupler position as shown in Figure 3-10b.
- 4 Draw construction arcs from point C_2 to O_4 and from point D_2 to O_4 to define the triangle $C_2O_4D_2$. This defines the relationship of the fixed pivot O_4 to the coupler line CD in the second coupler position as shown in Figure 3-10b.
- 5 Now transfer this relationship back to the first coupler position C_1D_1 so that the ground plane position $O_2'O_4'$ bears the same relationship to C_1D_1 as O_2O_4 bore to the second coupler po-

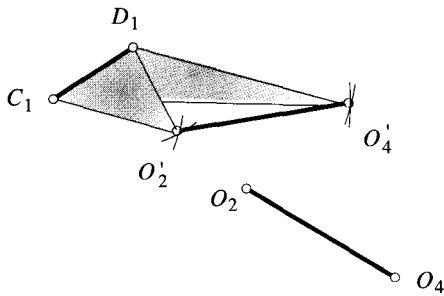
* This method and example were supplied by Mr. Homer D. Eckhardt, Consulting Engineer, Lincoln, MA.



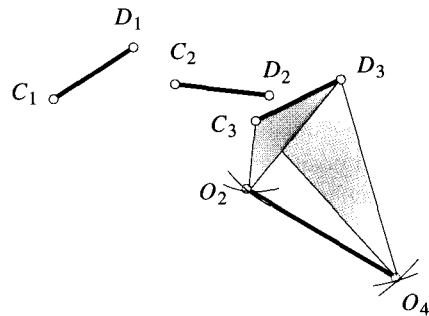
(a) Original coupler three-position problem with specified pivots



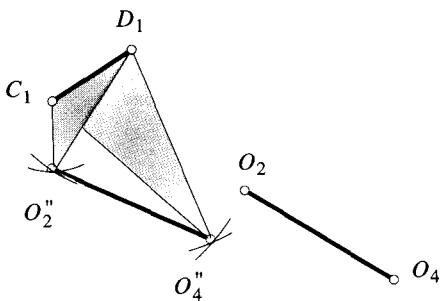
(b) Position of the ground plane relative to the second coupler position



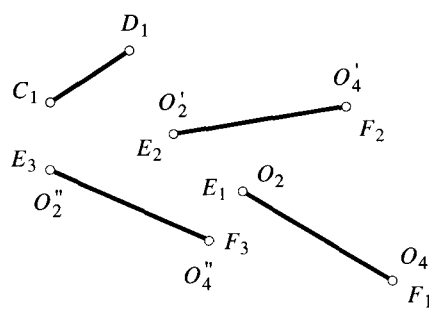
(c) Transferring second ground plane position to reference location at first position



(d) Position of the ground plane relative to the third coupler position



(e) Transferring third ground plane position to reference location at first position



(f) The three inverted positions of the ground plane corresponding to the original coupler position

FIGURE 3-10

Inverting the three-position motion synthesis problem

sition C_2D_2 . By doing this, we have pretended that the ground plane moved from O_2O_4 to $O_2'O_4'$ instead of the coupler moving from C_1D_1 to C_2D_2 . That is, we have *inverted* the problem.

- 6 Repeat the process for the third coupler position as shown in Figure 3-10d and transfer the third relative ground link position to the first, or reference, position as shown in Figure 3-10e.
- 7 The three inverted positions of the ground plane which correspond to the three desired coupler positions are labeled O_2O_4 , $O_2'O_4'$, and $O_2''O_4''$ and have also been renamed E_1F_1 , E_2F_2 , and E_3F_3 as shown in Figure 3-10f. These correspond to the three coupler positions shown in Figure 3-10a. Note that the original three lines C_1D_1 , C_2D_2 , and C_3D_3 are not now needed for the linkage synthesis.

We can use these three new lines E_1F_1 , E_2F_2 , and E_3F_3 to find the attachment points GH (moving pivots) on link 3 which will allow the desired fixed pivots O_2 and O_4 to be used for the three specified output positions. In effect we will now consider the ground link O_2O_4 to be a coupler moving through the inverse of the original three positions, find the “ground pivots” GH needed for that inverted motion, and put them on the real coupler instead. The inversion process done in Example 3-7 and Figure 3-10 has swapped the roles of coupler and ground plane. The remaining task is identical to that done in Example 3-5 and Figure 3-8. The result of the synthesis then must be reinverted to obtain the solution.

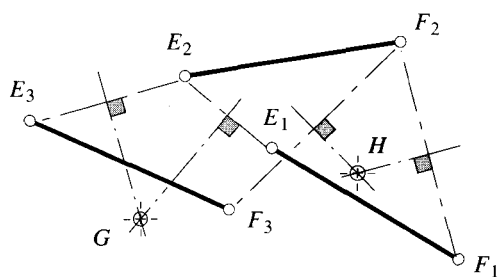
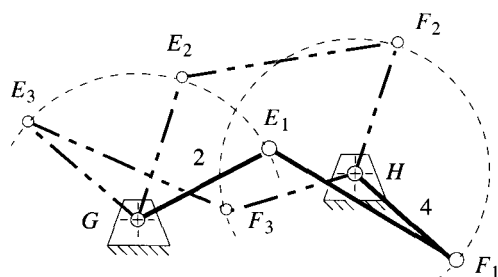
EXAMPLE 3-8

Finding the Moving Pivots for Three Positions and Specified Fixed Pivots.

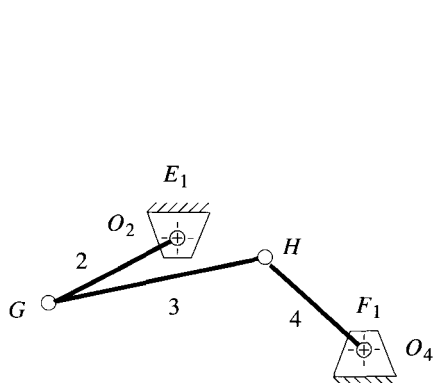
Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3 . Use specified fixed pivots O_2 and O_4 . Find the required moving pivot locations on the coupler by inversion.

Solution: Using the inverted ground link positions E_1F_1 , E_2F_2 , and E_3F_3 found in Example 3-7, find the fixed pivots for that inverted motion, then reinvert the resulting linkage to create the moving pivots for the three positions of coupler CD which use the selected fixed pivots O_2 and O_4 as shown in Figure 3-10a (see also Figure 3-11).

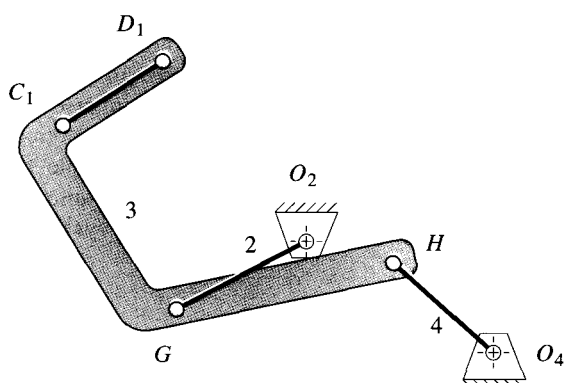
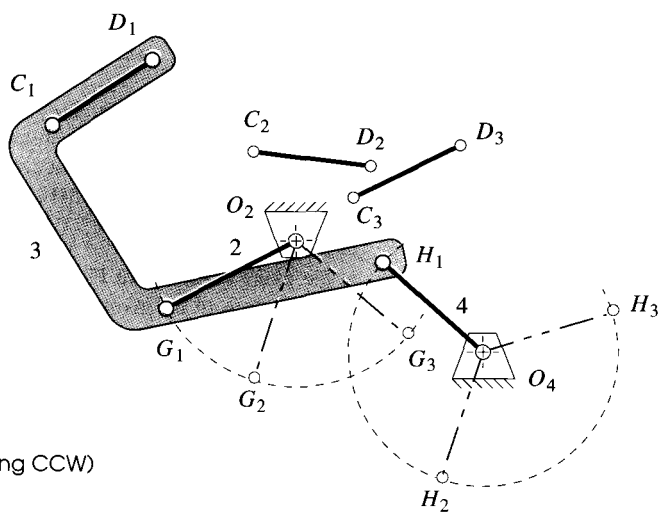
- 1 Start with the inverted three positions in the plane as shown in Figures 3-10f and 3-11a. Lines E_1F_1 , E_2F_2 , and E_3F_3 define the three positions of the inverted link to be moved.
- 2 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3 .
- 3 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection G .
- 4 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3 . Label the intersection H .
- 5 Connect G with E_1 and call it link 2. Connect H with F_1 and call it link 4. See Figure 3-11b.
- 6 In this inverted linkage, line E_1F_1 is the coupler, link 3. Line GH is the “ground” link 1.

(a) Construction to find rotapoles G and H 

(b) The correct inversion of desired linkage



(c) Reinvert to obtain the result

(d) Re-place line CD on link 3

(e) The three positions (link 4 driving CCW)

FIGURE 3-11

Constructing the linkage for three positions with specified fixed pivots by inversion

- 7 We must now reinvert the linkage to return to the original arrangement. Line E_1F_1 is really the ground link O_2O_4 , and GH is really the coupler. Figure 3-11c shows the reinversion of the linkage in which points G and H are now the moving pivots on the coupler and E_1F_1 has resumed its real identity as ground link O_2O_4 . (See Figure 3-10e.)
- 8 Figure 3-11d reintroduces the original line C_1D_1 in its correct relationship to line O_2O_4 at the initial position as shown in the original problem statement in Figure 3-10a. This forms the required coupler plane and defines a minimal shape of link 3.
- 9 The angular motions required to reach the second and third positions of line CD shown in Figure 3-11e are the same as those defined in Figure 3-11b for the linkage inversion. The angle F_1HF_2 in Figure 3-11b is the same as angle $H_1O_4H_2$ in Figure 3-11e and F_2HF_3 is the same as angle $H_2O_4H_3$. The angular excursions of link 2 retain the same relationship between Figure 3-11b and e as well. The angular motions of links 2 and 4 are the same for both inversions as the link excursions are relative to one another.
- 10 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case provided that the linkage has mobility among all three positions. This solution is a non-Grashof linkage.
- 11 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions. In this case links 3 and 4 reach a toggle position between points H_1 and H_2 . This means that this linkage cannot be driven from link 2 as it will hang up at that toggle position. It must be driven from link 4.

By inverting the original problem, we have reduced it to a more tractable form which allows a direct solution by the general method of three-position synthesis from Examples 3-5 and 3-6.

Position Synthesis for More Than Three Positions

It should be obvious that the more constraints we impose on these synthesis problems, the more complicated the task becomes to find a solution. When we define more than three positions of the output link, the difficulty increases substantially.

FOUR-POSITION SYNTHESIS does not lend itself as well to manual graphical solutions, though Hall [3] does present one approach. Probably the best approach is that used by Sandor and Erdman [4] and others, which is a quantitative synthesis method and requires a computer to execute it. Briefly, a set of simultaneous vector equations is written to represent the desired four positions of the entire linkage. These are then solved after some free choices of variable values are made by the designer. The computer program LINCAGES [1] by Erdman et al., and the program KINSYN [5] by Kaufman, both provide a convenient and user-friendly computer graphics based means to make the necessary design choices to solve the four-position problem. See Chapter 5 for further discussion.

3.5 QUICK-RETURN MECHANISMS

Many machine design applications have a need for a difference in average velocity between their "forward" and "return" strokes. Typically some external work is being done by the linkage on the forward stroke, and the return stroke needs to be accomplished as

rapidly as possible so that a maximum of time will be available for the working stroke. Many arrangements of links will provide this feature. The only problem is to synthesize the right one!

Fourbar Quick-Return

The linkage synthesized in Example 3-1 is perhaps the simplest example of a fourbar linkage design problem (see Figure 3-4, p. 84, and program FOURBAR disk file F03-04.4br). It is a crank-rocker which gives two positions of the rocker with equal time for the forward stroke and the return stroke. This is called a *non-quick-return* linkage, and it is a special case of the more general **quick-return** case. The reason for its non quick-return state is the positioning of the crank center O_2 on the chord B_1B_2 extended. This results in equal angles of 180 degrees being swept out by the crank as it drives the rocker from one extreme (toggle position) to the other. If the crank is rotating at constant angular velocity, as it will tend to when motor driven, then each 180 degree sweep, forward and back, will take the same time interval. Try this with your cardboard model from Example 3-1 by rotating the crank at uniform velocity and observing the rocker motion and velocity.

If the crank center O_2 is located off the chord B_1B_2 extended, as shown in Figure 3-1b (p. 80) and Figure 3-12, then unequal angles will be swept by the crank between the toggle positions (defined as colinearity of crank and coupler). Unequal angles will give unequal time, when the crank rotates at constant velocity. These angles are labeled α and β in Figure 3-12. Their ratio α/β is called the **time ratio** (T_R) and defines the degree of quick-return of the linkage. Note that the term **quick-return** is arbitrarily used to describe this kind of linkage. If the crank is rotated in the opposite direction, it will be a **quick-forward** mechanism. Given a completed linkage, it is a trivial task to estimate the time ratio by measuring or calculating the angles α and β . It is a more difficult task to design the linkage for a chosen time ratio. Hall^[6] provides a graphical method to synthesize a quick-return Grashof fourbar. To do so we need to compute the values of α and β which will give the specified time ratio. We can write two equations involving α and β and solve them simultaneously.

$$T_R = \frac{\alpha}{\beta} \qquad \alpha + \beta = 360 \qquad (3.1)$$

We also must define a construction angle,

$$\delta = |180 - \alpha| = |180 - \beta| \qquad (3.2)$$

which will be used to synthesize the linkage.

EXAMPLE 3-9

Fourbar Crank-Rocker Quick-Return Linkage for Specified Time Ratio.

Problem: Redesign Example 3-1 to provide a time ratio of 1:1.25 with 45° output rocker motion.

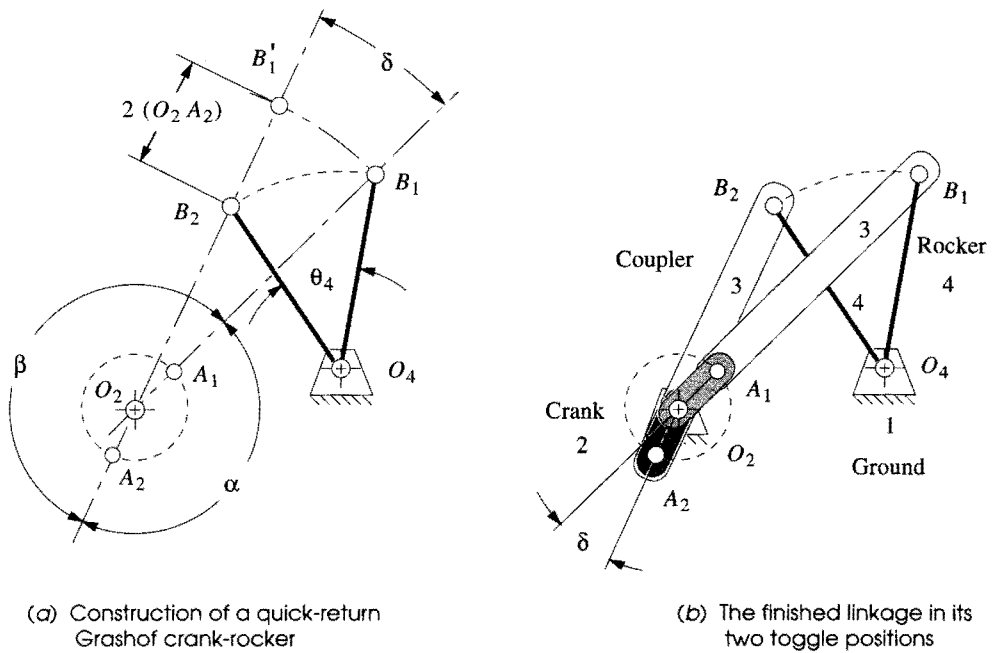


FIGURE 3-12

Quick-return Grashof fourbar crank-rocker linkage

Solution: (see Figure 3-12)

- 1 Draw the output link O_4B in both extreme positions, in any convenient location, such that the desired angle of motion, θ_4 , is subtended.
- 2 Calculate α , β , and δ using equations 3.1 and 3.2. In this example, $\alpha = 160^\circ$, $\beta = 200^\circ$, $\delta = 20^\circ$
- 3 Draw a construction line through point B_1 at any convenient angle.
- 4 Draw a construction line through point B_2 at angle δ from the first line.
- 5 Label the intersection of the two construction lines O_2 .
- 6 The line O_2O_4 now defines the ground link.
- 7 Calculate the lengths of crank and coupler by measuring O_2B_1 and O_2B_2 and solve simultaneously:

$$\text{Coupler} + \text{crank} = O_2B_1$$

$$\text{Coupler} - \text{crank} = O_2B_2$$

or you can construct the crank length by swinging an arc centered at O_2 from B_1 to cut line O_2B_2 extended. Label that intersection B_1' . The line B_2B_1' is twice the crank length. Bisect this line segment to measure crank length O_2A_1 .

- 8 Calculate the Grashof condition. If non-Grashof, repeat steps 3 to 8 with O_2 further from O_4 .

- 9 Make a cardboard model of the linkage and articulate it to check its function.
- 10 Check the transmission angles.

This method works well for time ratios down to about 1:1.5. Beyond that value the transmission angles become poor, and a more complex linkage is needed. Input the file F03-12.4br to program FOURBAR to see Example 3-9.

Sixbar Quick-Return

Larger time ratios, up to about 1:2, can be obtained by designing a sixbar linkage. The strategy here is to first design a fourbar drag link mechanism which has the desired time ratio between its driver crank and its driven or "dragged" crank, and then add a dyad (two-bar) output stage, driven by the dragged crank. This dyad can be arranged to have either a rocker or a translating slider as the output link. First the drag link fourbar will be synthesized; then the dyad will be added.

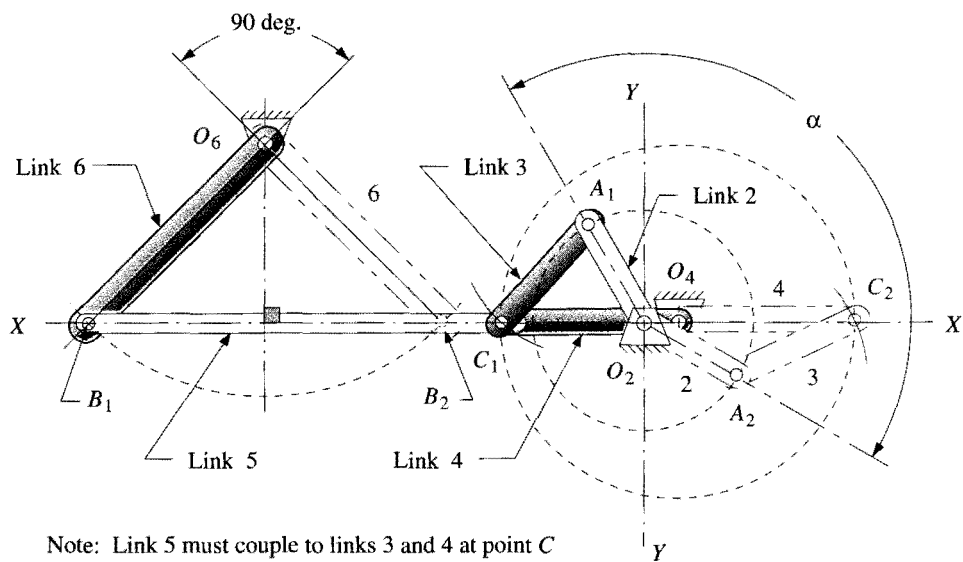
EXAMPLE 3-10

Sixbar Drag Link Quick-Return Linkage for Specified Time Ratio.

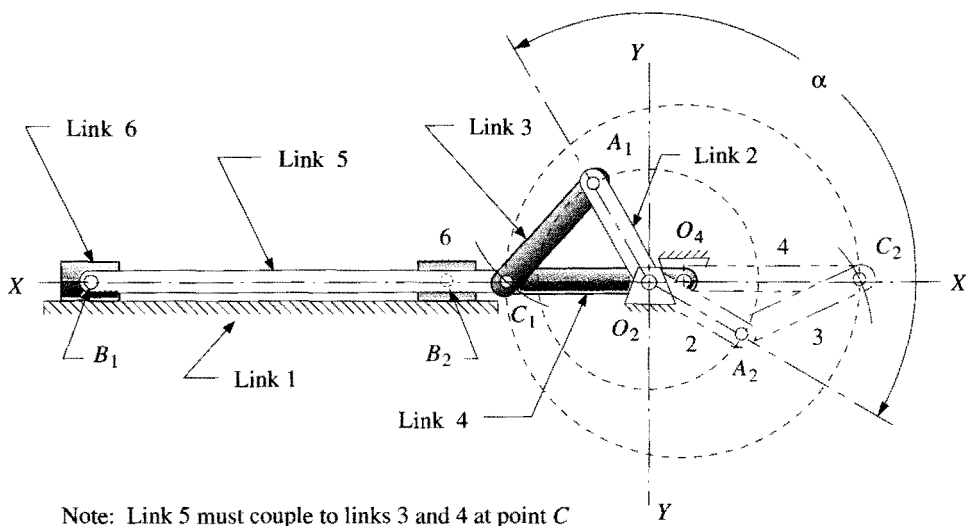
Problem: Provide a time ratio of 1:1.4 with 90° rocker motion.

Solution: (see Figure 3-13)

- 1 Calculate α and β using equations 3.1. For this example, $\alpha = 150^\circ$ and $\beta = 210^\circ$.
- 2 Draw a line of centers XX at any convenient location.
- 3 Choose a crank pivot location O_2 on line XX and draw an axis YY perpendicular to XX through O_2 .
- 4 Draw a circle of convenient radius O_2A about center O_2 .
- 5 Lay out angle α with vertex at O_2 , symmetrical about quadrant one.
- 6 Label points A_1 and A_2 at the intersections of the lines subtending angle α and the circle of radius O_2A .
- 7 Set the compass to a convenient radius AC long enough to cut XX in two places on either side of O_2 when swung from both A_1 and A_2 . Label the intersections C_1 and C_2 .
- 8 The line O_2A_1 is the driver crank, link 2, and line A_1C_1 is the coupler, link 3.
- 9 The distance C_1C_2 is twice the driven (dragged) crank length. Bisect it to locate the fixed pivot O_4 .
- 10 The line O_2O_4 now defines the ground link. Line O_4C_1 is the driven crank, link 4.
- 11 Calculate the Grashof condition. If non-Grashof, repeat steps 7 to 11 with a shorter radius in step 7.



(a) Rocker output sixbar drag link quick-return mechanism



(b) Slider output sixbar drag link quick-return mechanism

FIGURE 3-13

Synthesizing a sixbar drag link quick-return mechanism

- 12 Invert the method of Example 3-1 to create the output dyad using XX as the chord and O_4C_1 as the driving crank. The points B_1 and B_2 will lie on line XX and be spaced apart a distance $2O_4C_1$. The pivot O_6 will lie on the perpendicular bisector of B_1B_2 , at a distance from line XX which subtends the specified output rocker angle.
- 13 Check the transmission angles.

This linkage provides a quick-return when a constant-speed motor is attached to link 2. Link 2 will go through angle α while link 4 (which is dragging the output dyad along) goes through the first 180 degrees, from position C_1 to C_2 . Then, while link 2 completes its cycle through β degrees, the output stage will complete another 180 degrees from C_2 to C_1 . Since angle β is greater than α , the forward stroke takes longer. Note that the chordal stroke of the output dyad is twice the crank length O_4C_1 . This is independent of the angular displacement of the output link which can be tailored by moving the pivot O_6 closer to or further from the line XX .

The transmission angle at the joint between link 5 and link 6 will be optimized if the fixed pivot O_6 is placed on the perpendicular bisector of the chord B_1B_2 as shown in Figure 3-13a (p. 101). If a translating output is desired, the slider (link 6) will be located on line XX and will oscillate between B_1 and B_2 as shown in Figure 3-13b. The arbitrarily chosen size of this or any other linkage can be scaled up or down, simply by multiplying all link lengths by the same scale factor. Thus a design made to arbitrary size can be fit to any package. Input the file F03-13a.6br to program SIXBAR to see Example 3-10 in action.

CRANK-SLIDER QUICK-RETURN A commonly used mechanism, capable of large time ratios is shown in Figure 3-14. It is often used in metal shaper machines to provide

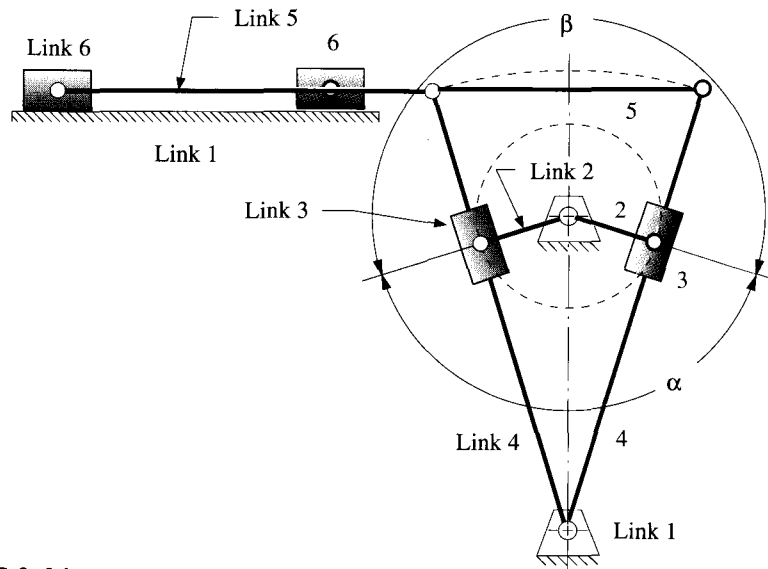


FIGURE 3-14

Crank-shaper quick-return mechanism

a slow cutting stroke and a quick-return stroke when the tool is doing no work. It is the inversion #2 of the slider-crank mechanism as was shown in Figure 2-13b (p. 44). It is very easy to synthesize this linkage by simply moving the rocker pivot O_4 along the vertical centerline O_2O_4 while keeping the two extreme positions of link 4 tangent to the circle of the crank, until the desired time ratio (α/β) is achieved. Note that the angular displacement of link 4 is then defined as well. Link 2 is the input and link 6 is the output.

Depending on the relative lengths of the links this mechanism is known as a **Whitworth** or **crank-shaper** mechanism. If the ground link is the shortest, then it will behave as a double-crank linkage, or *Whitworth mechanism*, with both pivoted links making full revolutions as shown in Figure 2-13b (p. 44). If the driving crank is the shortest link, then it will behave as a crank-rocker linkage, or *crank-shaper mechanism*, as shown in Figure 3-14. They are the same inversion as the slider block is in complex motion in each case.

3.6 COUPLER CURVES

A **coupler** is the most interesting link in any linkage. It is in complex motion, and thus points on the coupler can have path motions of high degree.* In general, the more links, the higher the degree of curve generated, where degree here means *the highest power of any term in its equation*. A curve (function) can have up to as many intersections (roots) with any straight line as the degree of the function. The *fourbar slider-crank* has, in general, fourth-degree coupler curves; the *pin-jointed fourbar*, up to sixth degree.† The geared fivebar, the sixbar, and more complicated assemblies will have still higher degree curves. Wunderlich [7b] derived an expression for the highest degree m possible for a coupler curve of a mechanism of n links connected with only revolute joints.

$$m = 2 \cdot 3^{(n/2-1)} \quad (3.3)$$

This gives, respectively, degrees of 6, 18, and 54 for the fourbar, sixbar, and eightbar linkage coupler curves. Specific points on their couplers may have degenerate curves of lower degree as, for example, the pin joints between any crank or rocker and the coupler that describes second-degree curves (circles). The parallelogram fourbar linkage has degenerate coupler curves, all of which are circles.

All linkages that possess one or more “floating” coupler links will generate coupler curves. It is interesting to note that these will be closed curves even for non-Grashof linkages. The coupler (or any link) can be extended infinitely in the plane. Figure 3-15 shows a fourbar linkage with its coupler extended to include a large number of points, each of which describes a different coupler curve. Note that these points may be anywhere on the coupler, including along line AB . There is, of course, an infinity of points on the coupler, each of which generates a different curve.

Coupler curves can be used to generate quite useful path motions for machine design problems. They are capable of *approximating straight lines* and *large circle arcs* with remote centers. Recognize that the coupler curve is a solution to the path generation problem described in Section 3.2 (p. 78). It is not by itself a solution to the motion generation problem, since the attitude or orientation of a line on the coupler is not predicted by the information contained in the path. Nevertheless it is a very useful device, and it can be converted to a parallel motion generator by adding two links as described

* In 1876, Kempe [7a] proved his theory that a linkage with only revolute (pin) and prismatic (slider) joints can be found that will trace any algebraic curve of any order or complexity. But the linkage for a particular curve may be excessively complex, may be unable to traverse the curve without encountering limit (toggle) positions, and may even need to be disassembled and reassembled to reach all points on the curve. See the discussion of circuit and branch defects in Section 4.12 (p. 173). Nevertheless this theory points to the potential for interesting motions from the coupler curve.

† The algebraic equation of the coupler curve is sometimes referred to as a “tricircular sextic” referring respectively to its circularity of 3 and its degree of 6.